```
> res := evalf(25^(1/8))
```

$$res := 1.495348781 \quad (1)$$

```
> res^8
```

$$24.99999997 \quad (2)$$

```
> f1 := x->x·(1-x)
```

$$f1 := x \mapsto x \cdot (1 - x) \quad (3)$$

```
> solve(f1(x) = x)
```

$$0,\ 0 \quad (4)$$

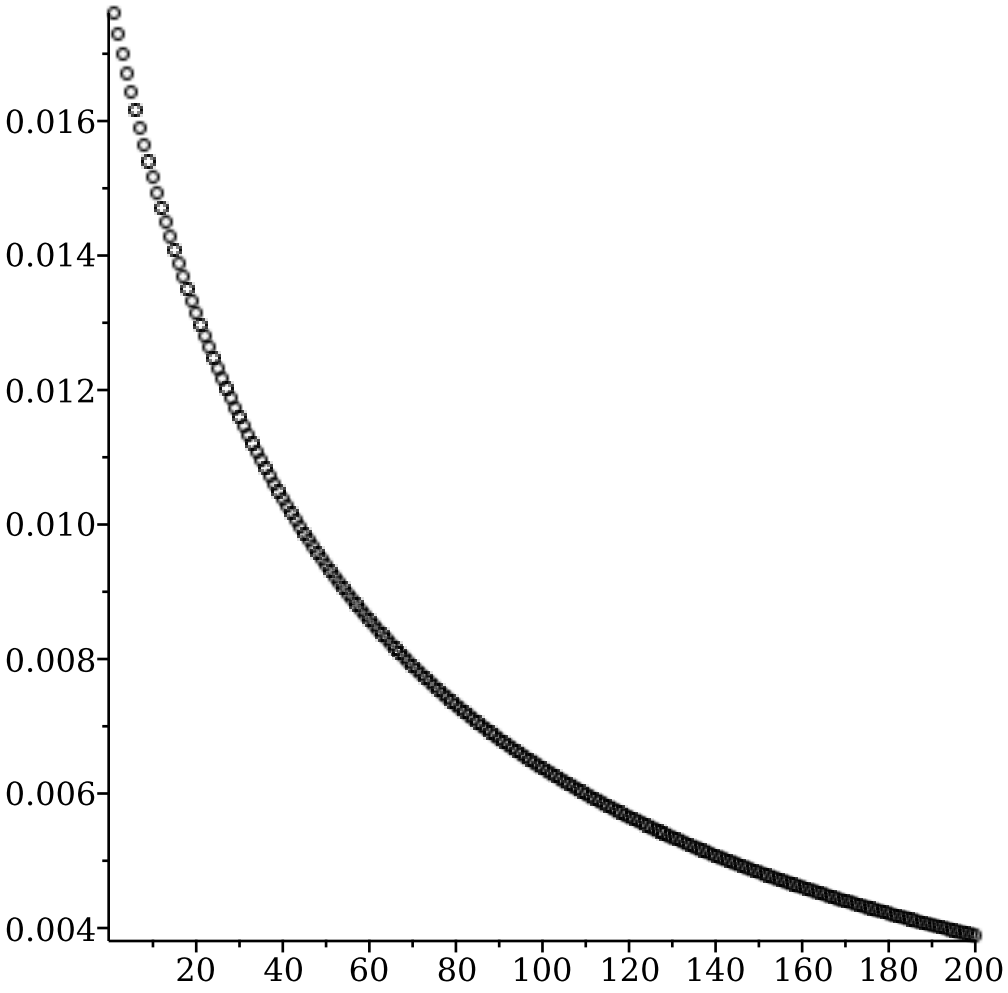
```
> x := 0.5
```

$$x := 0.5 \quad (5)$$

```
> for i from 1 to 200 do x := f1(x) : psi(i) := x;od:
> with(plots)
```

[*animate, animate3d, animatecurve, arrow, changecoords, complexplot, complexplot3d, conformal, conformal3d, contourplot, contourplot3d, coordplot, coordplot3d, densityplot, display, dualaxisplot, fieldplot, fieldplot3d, gradplot, gradplot3d, implicitplot, implicitplot3d, inequal, interactive, interactiveparams, intersectplot, listcontplot, listcontplot3d, listdensityplot, listplot, listplot3d, loglogplot, logplot, matrixplot, multiple, odeplot, pareto, plotcompare, pointplot, pointplot3d, polarplot, polygonplot, polygonplot3d, polyhedra_supported, polyhedraplot, rootlocus, semilogplot, setcolors, setoptions, setoptions3d, shadebetween, spacecurve, sparsematrixplot, surfdata, textplot, textplot3d, tubeplot*] (6)

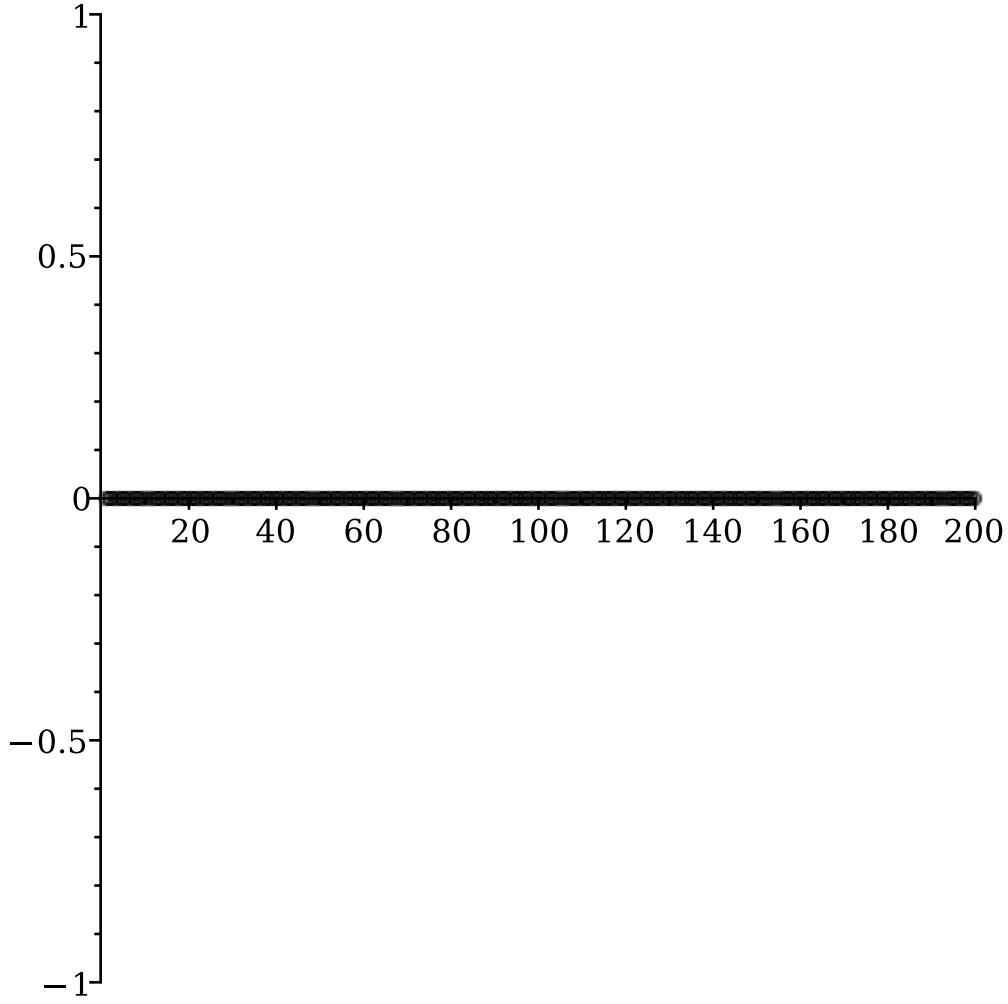
```
> points := [[n, psi(n)]$n = 1..200] : pointplot(points, symbol = circle)
```




```
> x := 0
```

$$x := 0 \qquad (7)$$

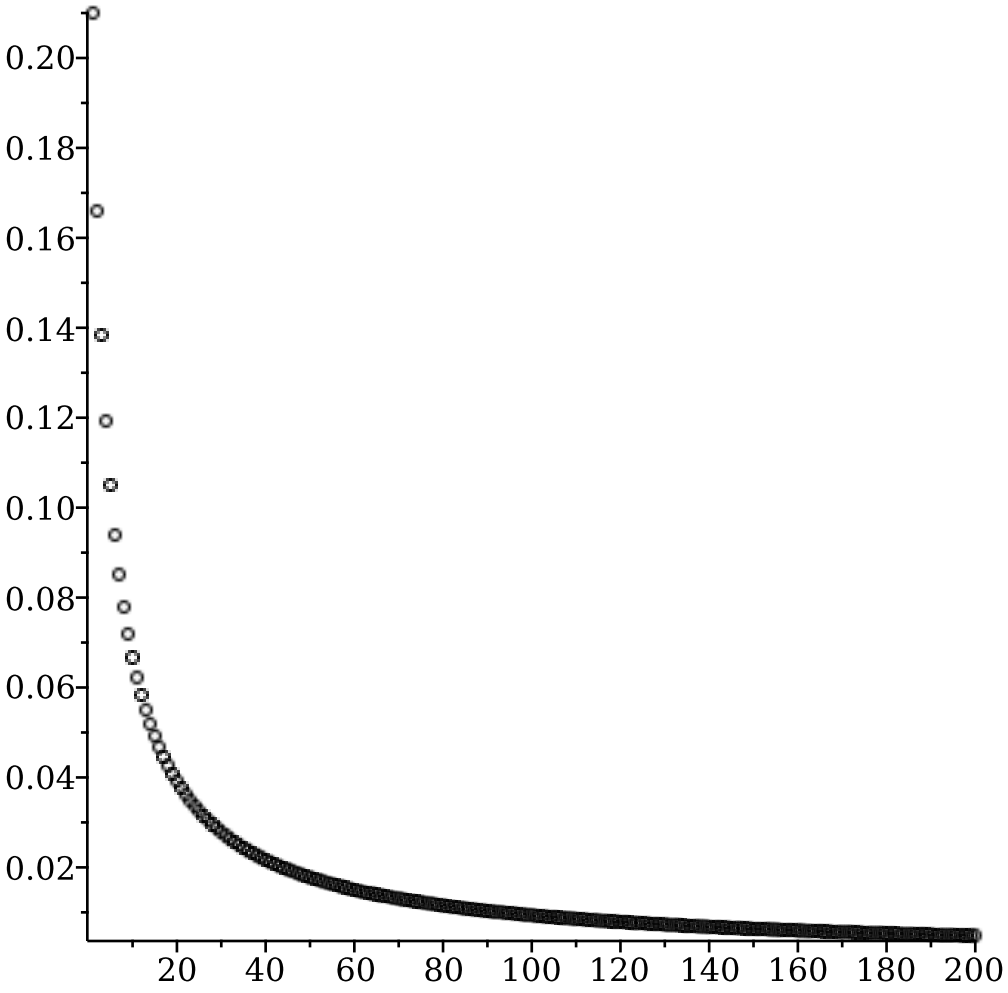
```
> for i from 1 to 200 do x := f1(x) : psi(i) := x;od:
> points := [[n, psi(n)]$n = 1..200] : with(plots) : pointplot(points, symbol
      = circle)
```




```
> x := 0.7
```

$$x := 0.7 \qquad (8)$$

```
> for i from 1 to 200 do x := f1(x) : psi(i) := x;od:
> points := [[n, psi(n)]$n = 1..200] : with(plots) : pointplot(points, symbol
      = circle)
```




```
> x := 'x'
```

$$x := x \tag{9}$$

```
> f2 := x→2·x·(1−x)
```

$$f2 := x \mapsto 2 \cdot x \cdot (1 - x) \tag{10}$$

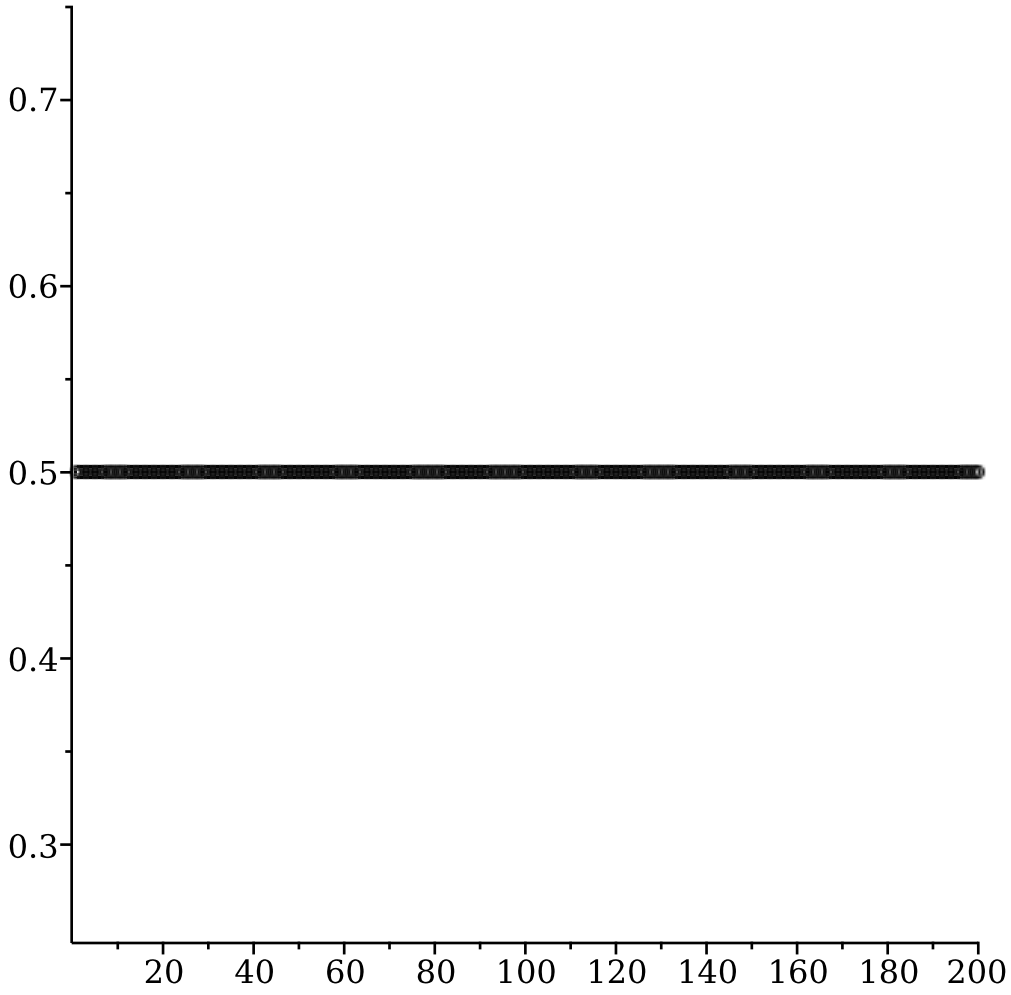
```
> solve(f2(x) = x, x)
```

$$0,\ \frac{1}{2} \tag{11}$$

```
> x := 0.5
```

$$x := 0.5 \tag{12}$$

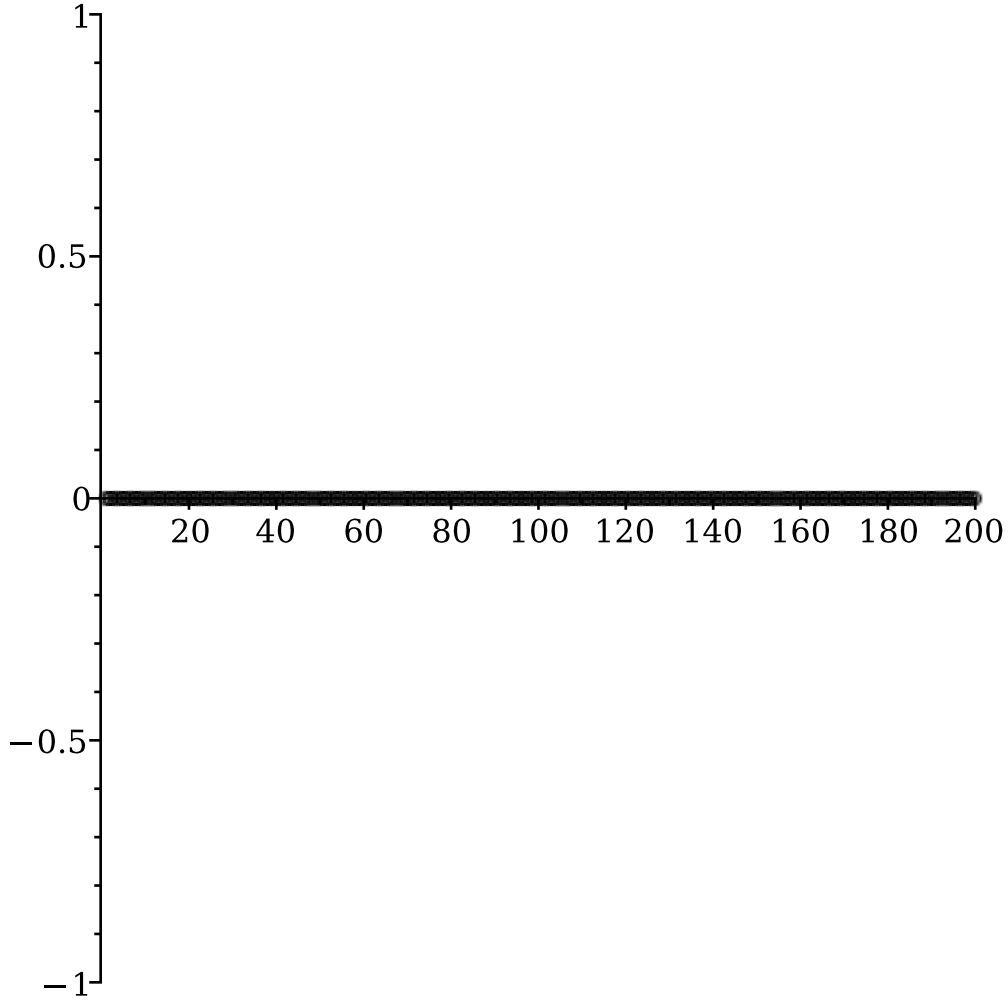
```
> for i from 1 to 200 do x := f2(x) : psi(i) := x;od:
> points := [[n, psi(n)]$n = 1..200] : with(plots) : pointplot(points, symbol
      = circle)
```




```
> x := 0
```

$$x := 0 \qquad (13)$$

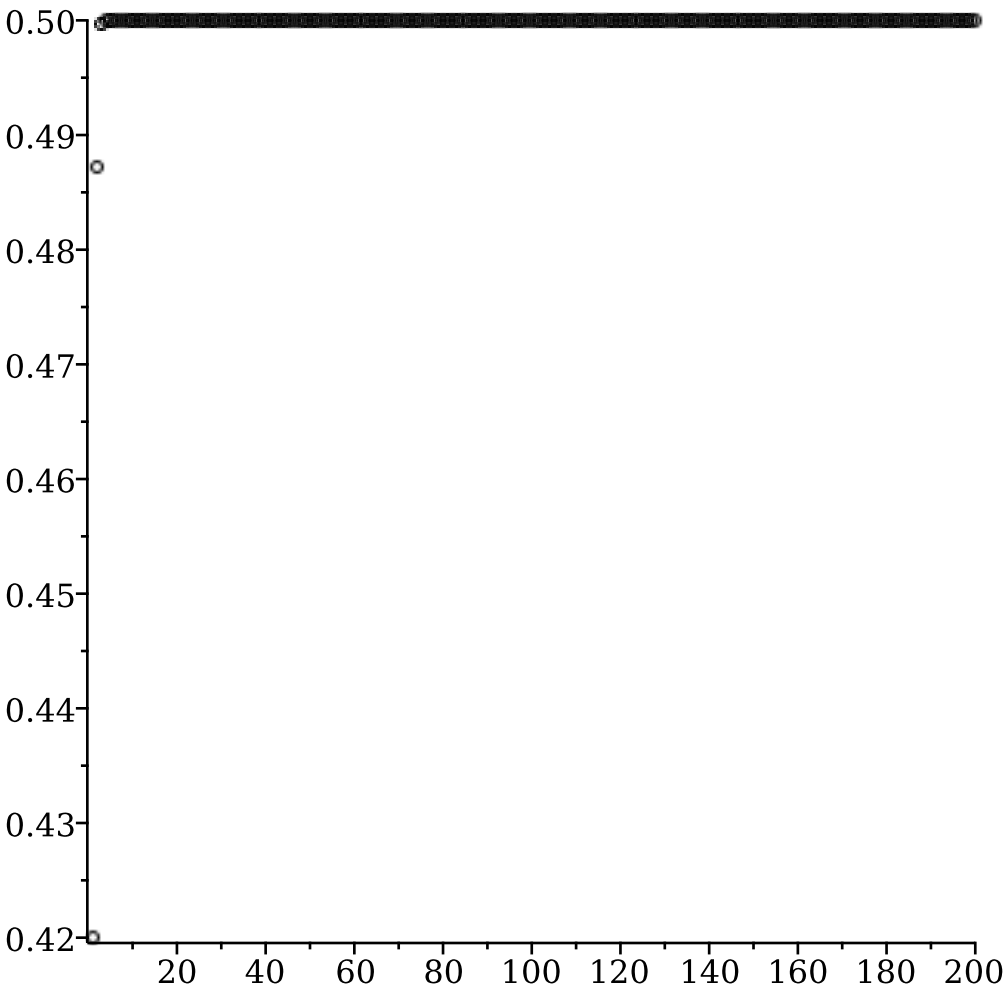
```
> for i from 1 to 200 do x := f2(x) : psi(i) := x;od:
> points := [[n, psi(n)]$n = 1..200] : with(plots) : pointplot(points, symbol
      = circle)
```




```
> x := 0.7
```

$$x := 0.7 \qquad (14)$$

```
> for i from 1 to 200 do x := f2(x) : psi(i) := x;od:
> points := [[n, psi(n)]$n = 1..200] : with(plots) : pointplot(points, symbol
      = circle)
```




```
> x := 'x'
```

$$x := x \qquad (15)$$

```
> f3 := x->3.1*x*(1-x)
```

$$f3 := x \mapsto 3.1 \cdot x \cdot (1 - x) \qquad (16)$$

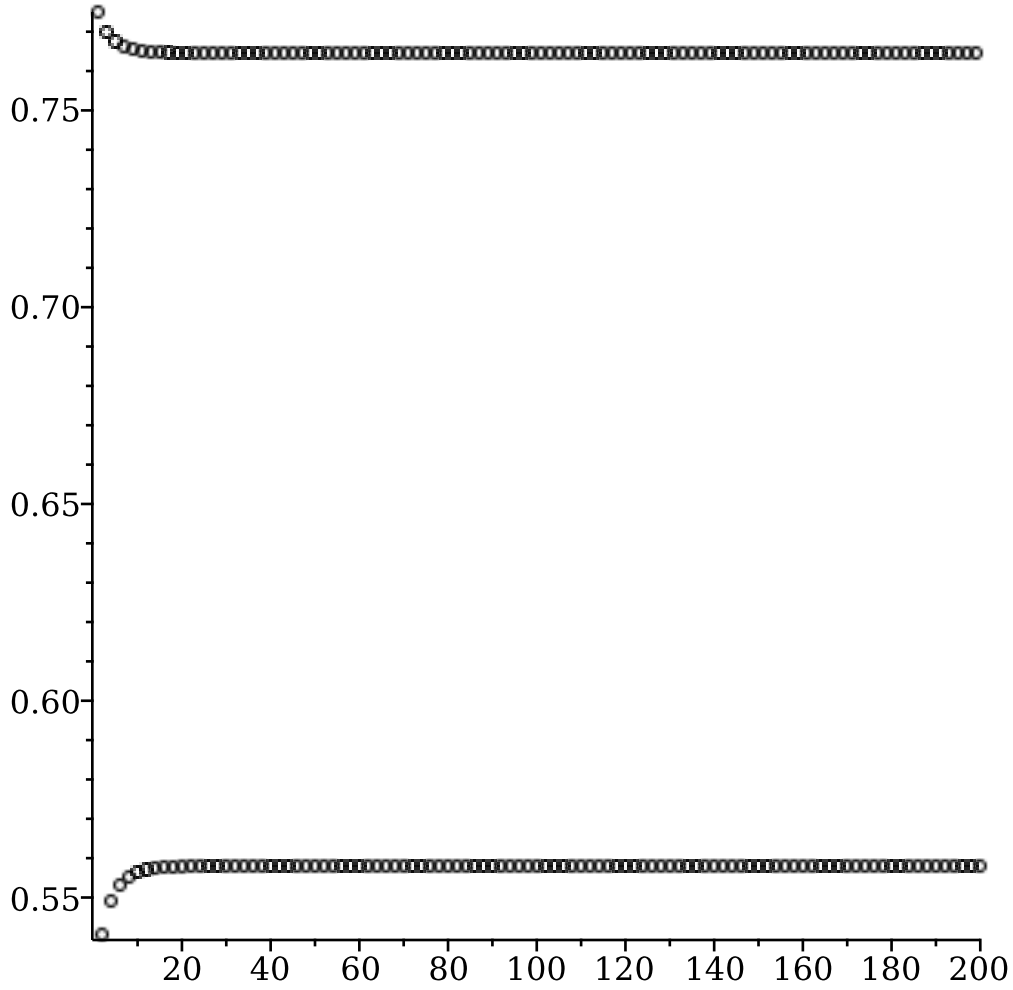
```
> solve(f3(x) = x, x)
```

$$0., 0.6774193548 \qquad (17)$$

```
> x := 0.5
```

$$x := 0.5 \qquad (18)$$

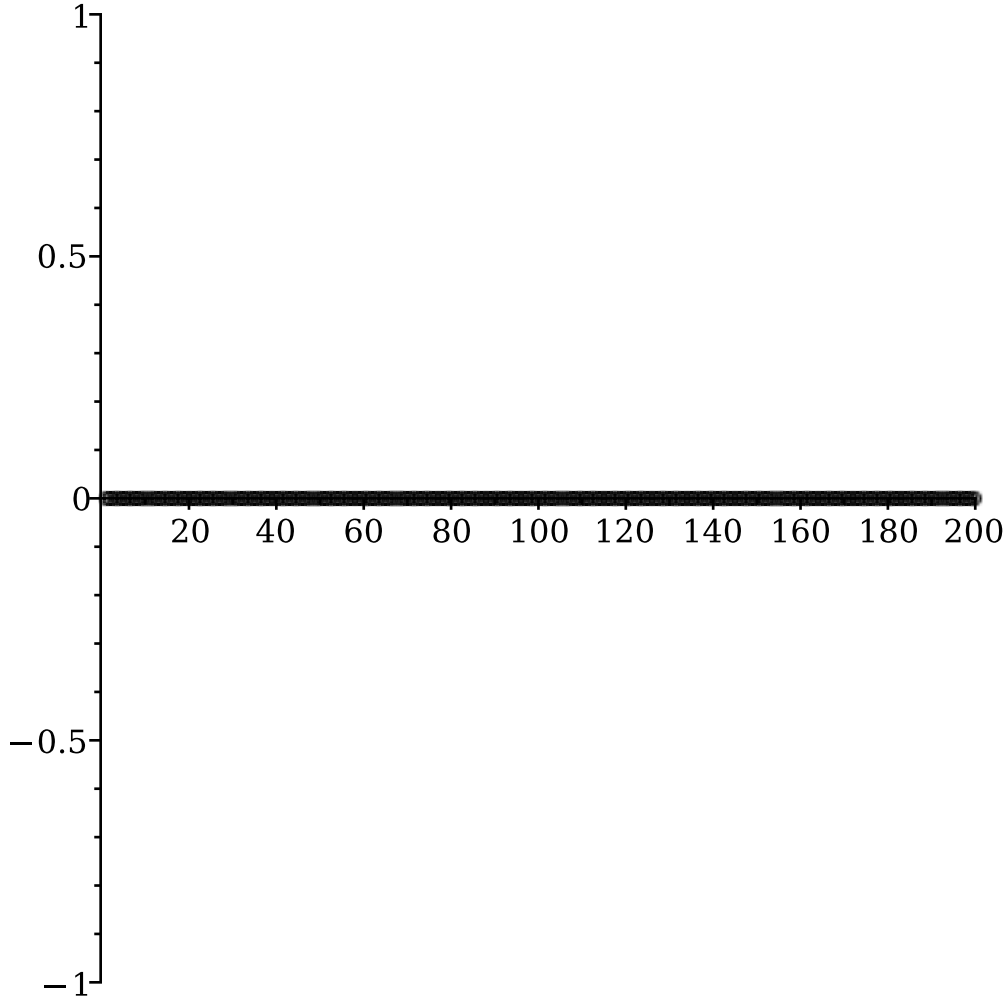
```
>  for i from 1 to 200 do x := f3(x) : psi(i) := x;od:
> points := [[n, psi(n)]$n = 1..200] : with(plots) : pointplot(points, symbol
    = circle)
```




```
> x := 0
```

$$x := 0 \qquad (19)$$

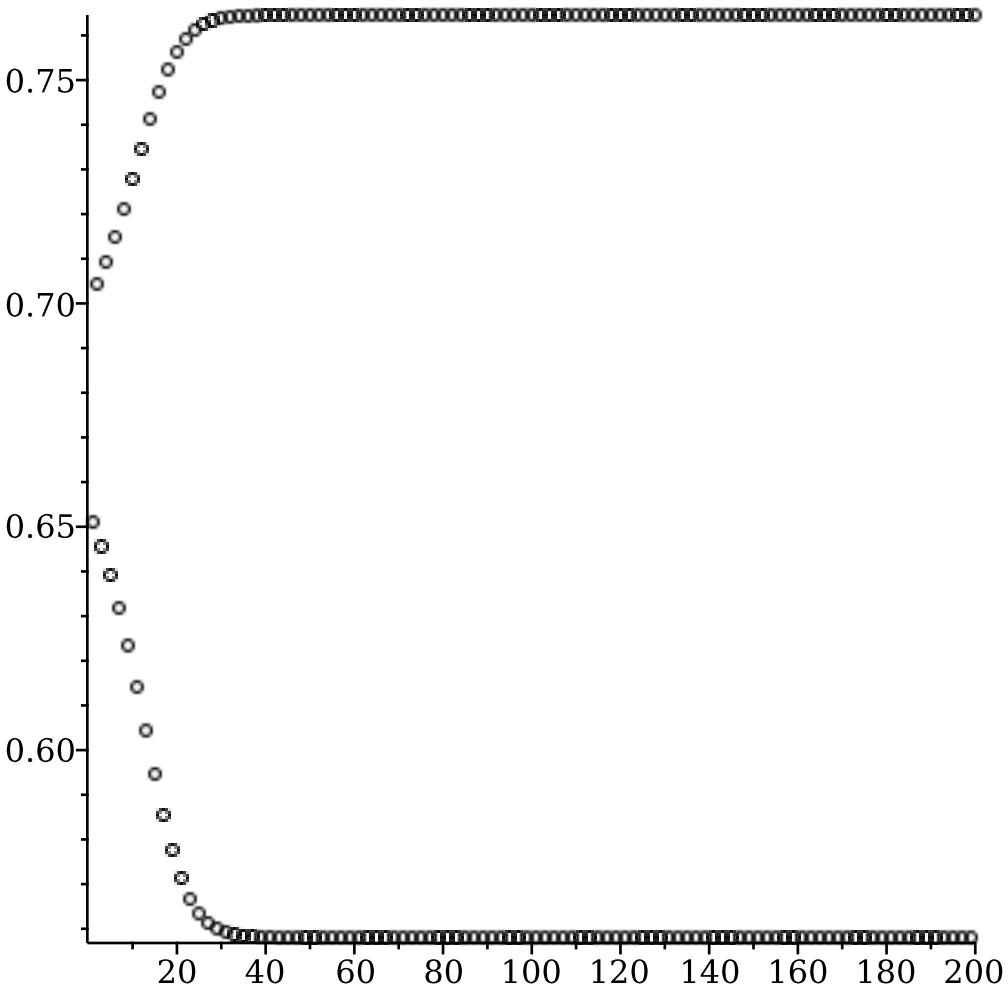
```
> for i from 1 to 200 do x := f3(x) : psi(i) := x;od:
> points := [[n, psi(n)]$n = 1..200] : with(plots) : pointplot(points, symbol
      = circle)
```




```
> x := 0.7
```

$$x := 0.7 \qquad (20)$$

```
> for i from 1 to 200 do x := f3(x) : psi(i) := x;od:
> points := [[n, psi(n)]$n = 1..200] : with(plots) : pointplot(points, symbol
      = circle)
```




```
> x := 'x'
```

$$x := x \qquad (21)$$

```
> f4 := x->3.5*x*(1-x)
```

$$f4 := x \mapsto 3.5 \cdot x \cdot (1 - x) \qquad (22)$$

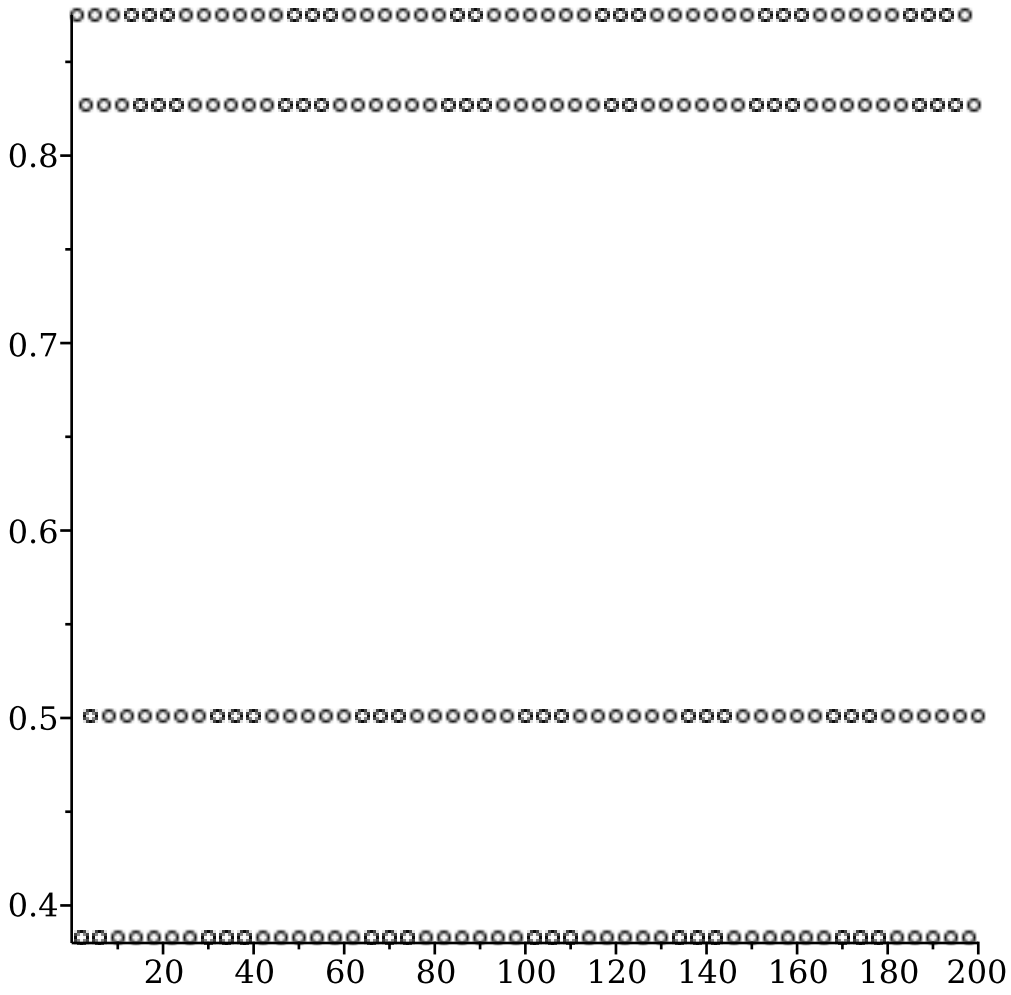
```
> solve(f4(x) = x, x)
```

$$0., 0.7142857143 \qquad (23)$$

```
> x := 0.5
```

$$x := 0.5 \qquad (24)$$

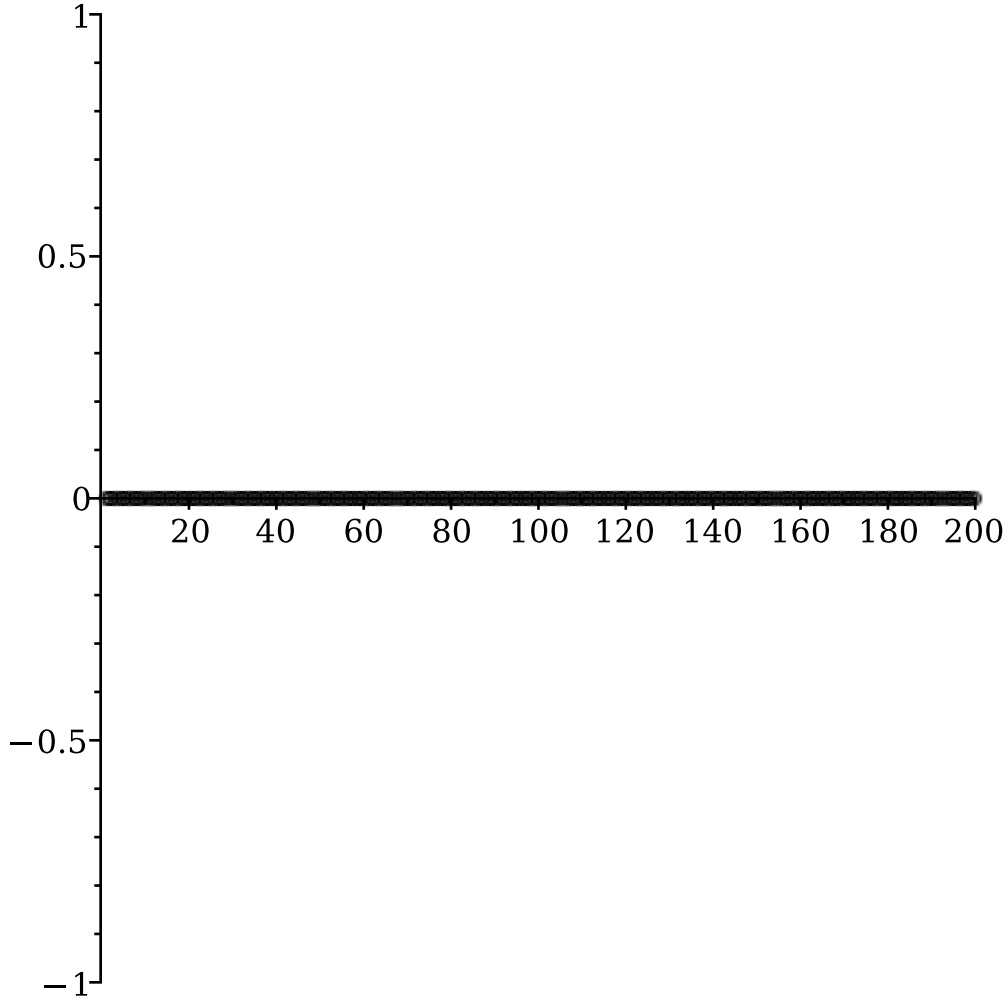
```
> for i from 1 to 200 do x := f4(x) : psi(i) := x;od:
> points := [[n, psi(n)]$n = 1..200] : with(plots) : pointplot(points, symbol
      = circle)
```




```
> x := 0
```

$$x := 0 \qquad (25)$$

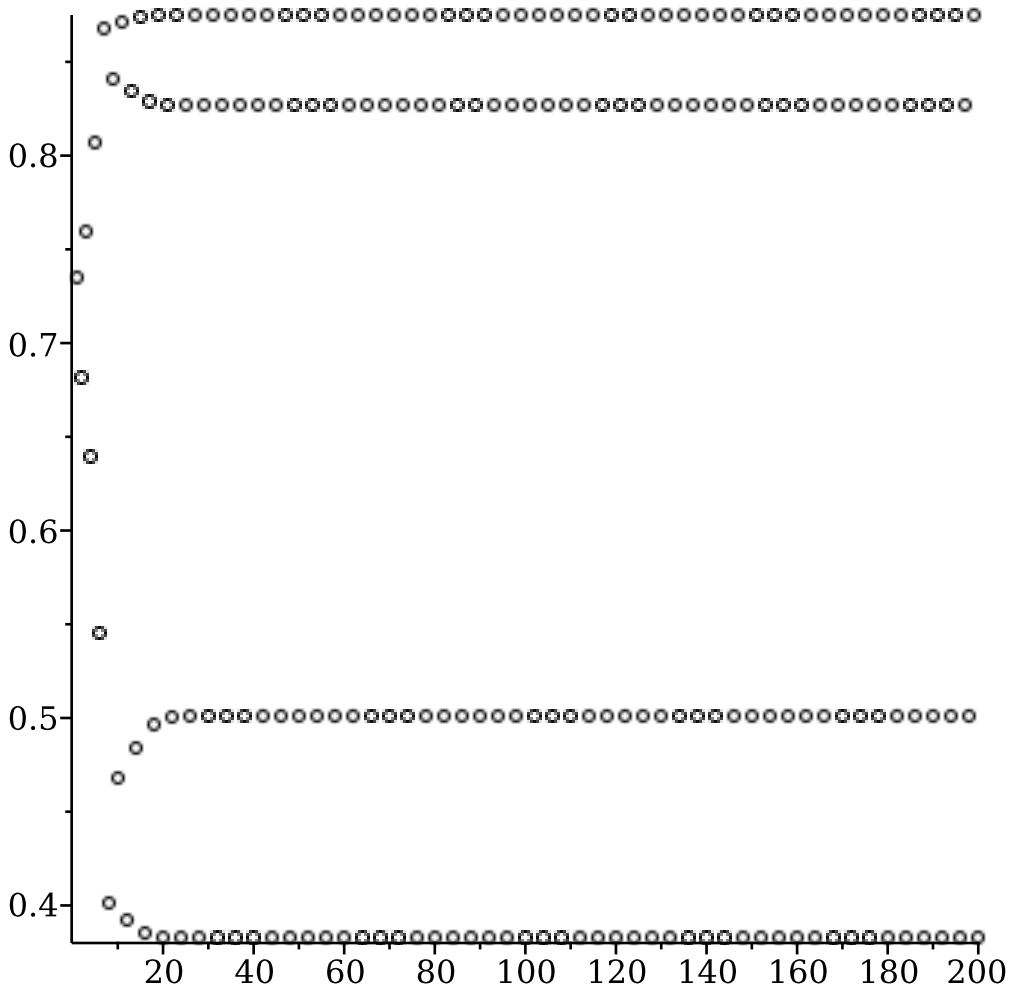
```
> for i from 1 to 200 do x := f4(x) : psi(i) := x;od:
> points := [[n, psi(n)]$n = 1..200] : with(plots) : pointplot(points, symbol
      = circle)
```




```
> x := 0.7
```

$$x := 0.7 \qquad (26)$$

```
> for i from 1 to 200 do x := f4(x) : psi(i) := x;od:
> points := [[n, psi(n)]$n = 1..200] : with(plots) : pointplot(points, symbol
      = circle)
```



> $f5 := x \to 3.55 \cdot x \cdot (1 - x)$

$$f5 := x \mapsto 3.55 \cdot x \cdot (1 - x) \quad (27)$$

> $x := 'x'$

$$x := x \quad (28)$$

> $solve(f5(x) = x, x)$

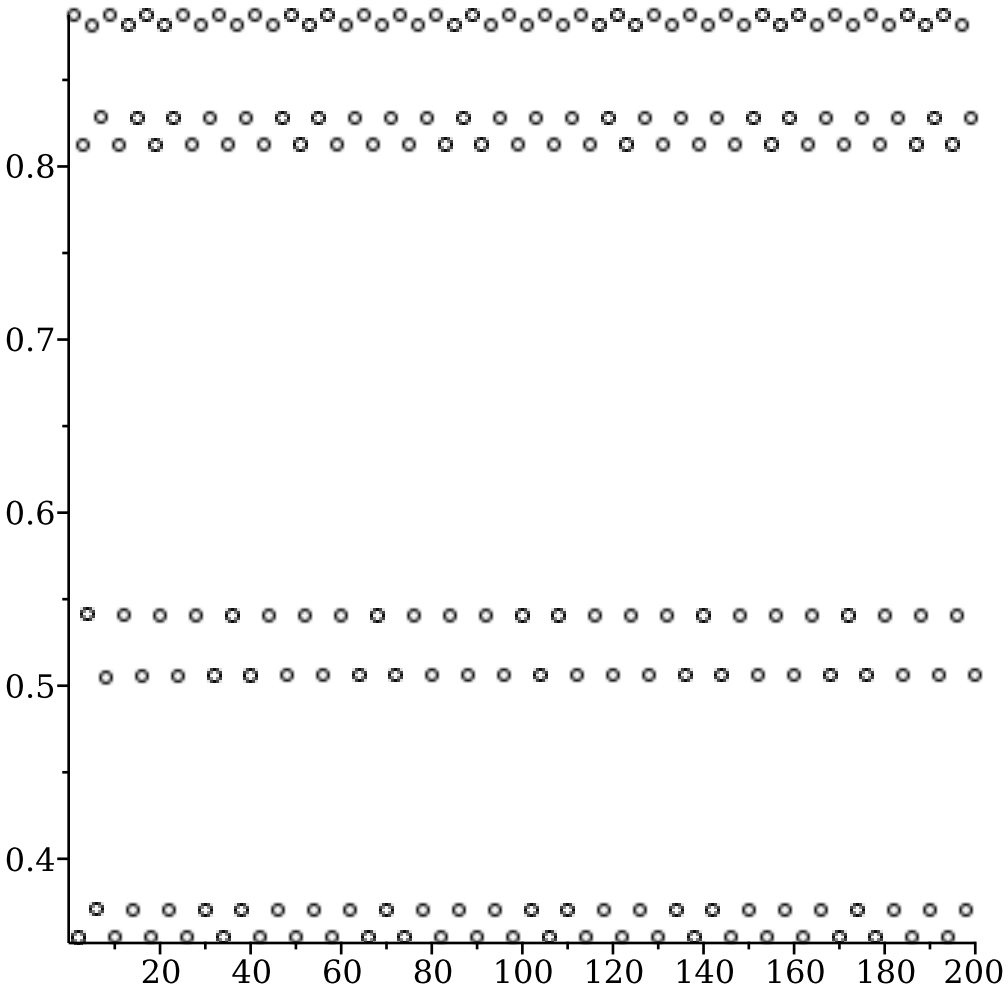
$$0., 0.7183098592 \quad (29)$$

> $x := 0.5$

$$x := 0.5 \quad (30)$$

> **for** i **from** 1 **to** 200 **do** $x := f5(x) : \mathrm{psi}(i) := x$;**od**:

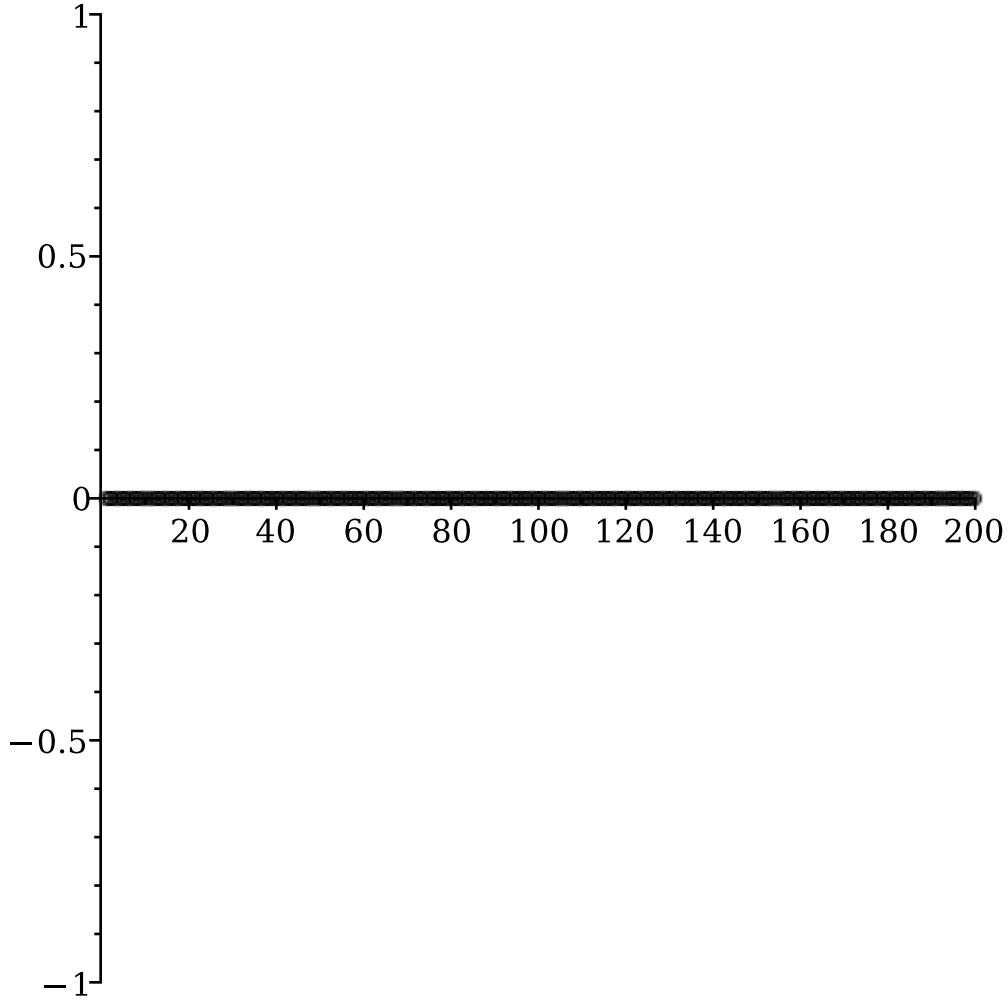
> $points := [[n, \mathrm{psi}(n)]\$n = 1..200] : with(plots) : pointplot(points, symbol = circle)$




```
> x := 0
```

$$x := 0 \qquad (31)$$

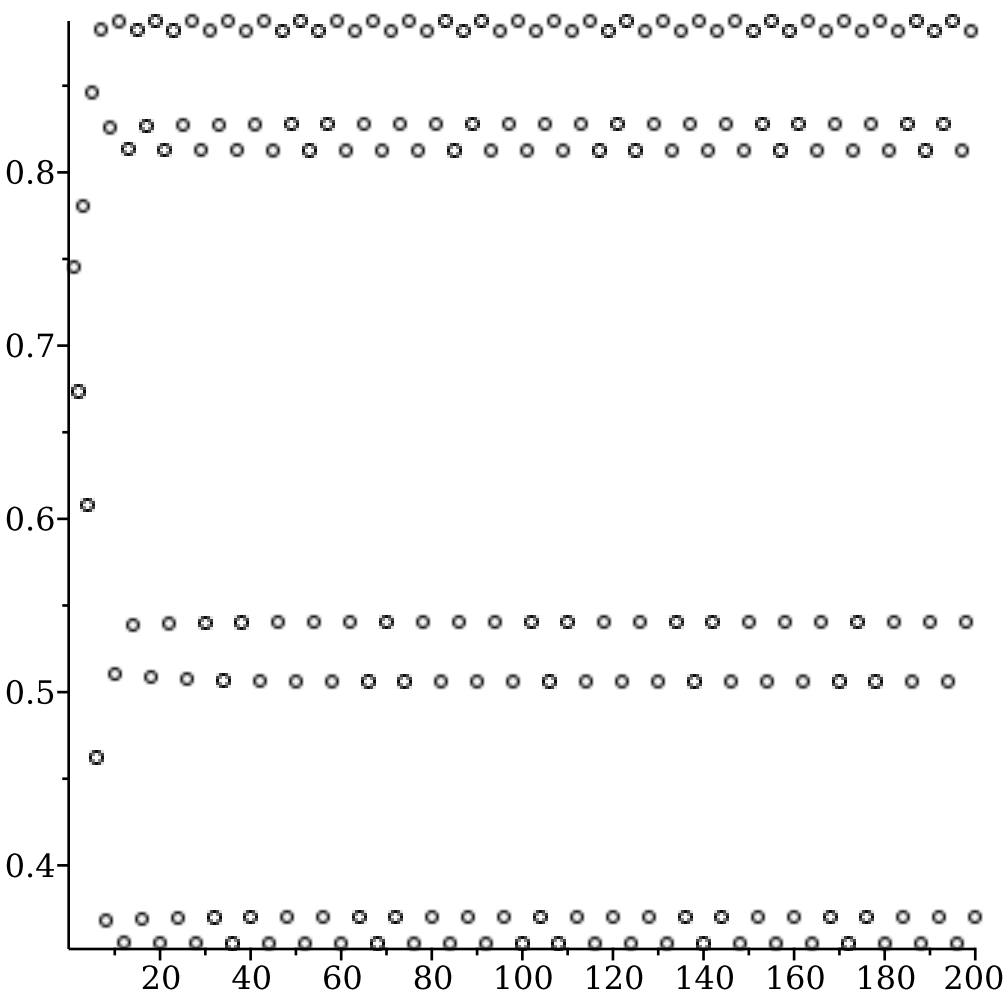
```
> for i from 1 to 200 do x := f5(x) : psi(i) := x;od:
> points := [[n, psi(n)]$n = 1..200] : with(plots) : pointplot(points, symbol
      = circle)
```




```
> x := 0.7
```

$$x := 0.7 \qquad (32)$$

```
> for i from 1 to 200 do x := f5(x) : psi(i) := x;od:
> points := [[n, psi(n)]$n = 1..200] : with(plots) : pointplot(points, symbol
      = circle)
```




```
> f6 := x→3.6·x·(1 − x)
```

$$f6 := x \mapsto 3.6 \cdot x \cdot (1 - x) \quad (33)$$

```
> x :='x'
```

$$x := x \quad (34)$$

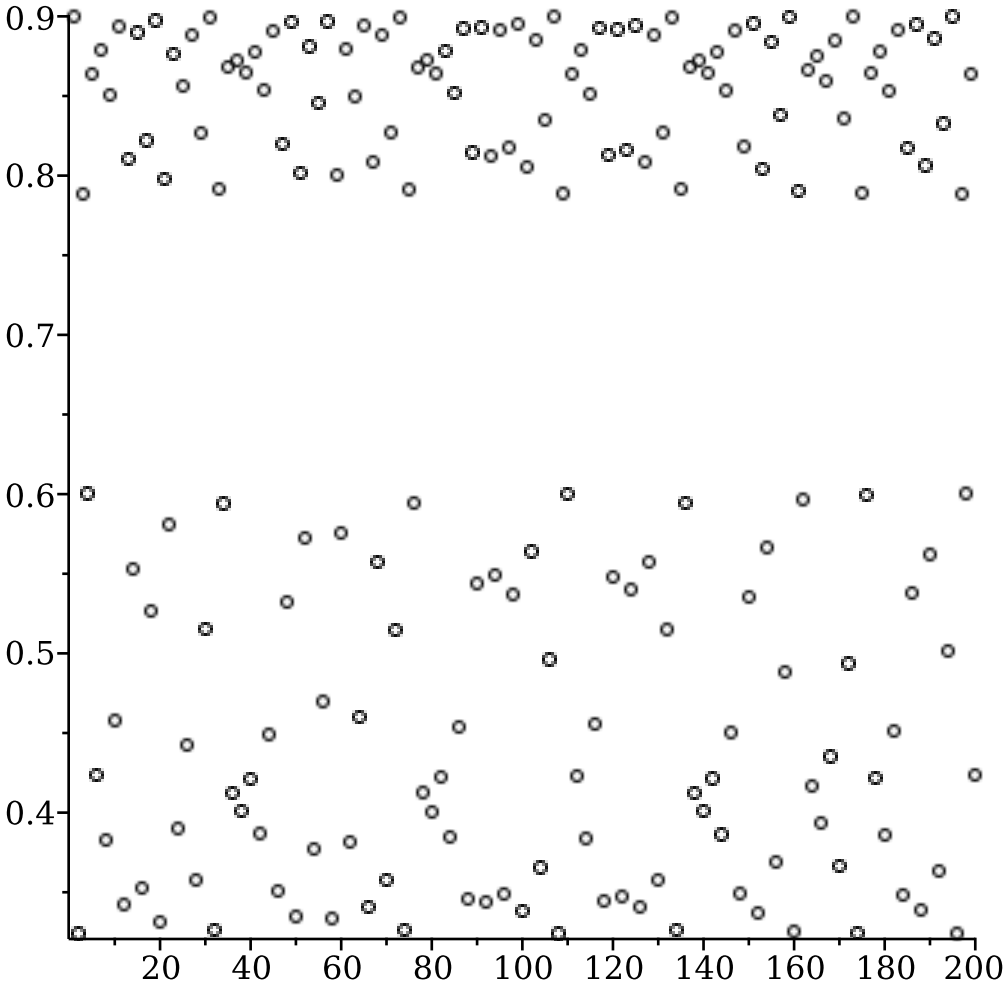
```
> solve(f6(x) = x, x)
```

$$0., 0.7222222222 \quad (35)$$

```
> x := 0.5
```

$$x := 0.5 \quad (36)$$

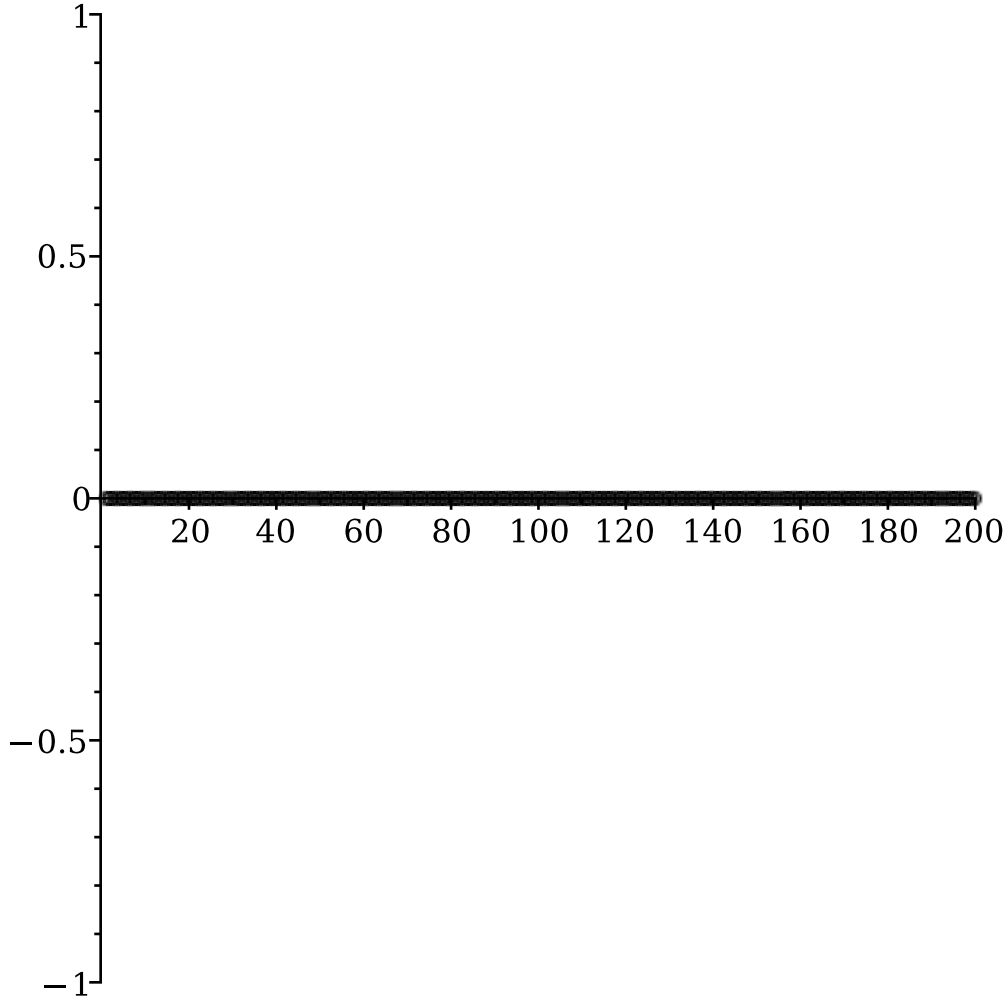
```
> for i from 1 to 200 do x := f6(x) : psi(i) := x;od:
> points := [[n, psi(n)]$n = 1..200] : with(plots) : pointplot(points, symbol
      = circle)
```




```
> x := 0
```

$$x := 0 \qquad (37)$$

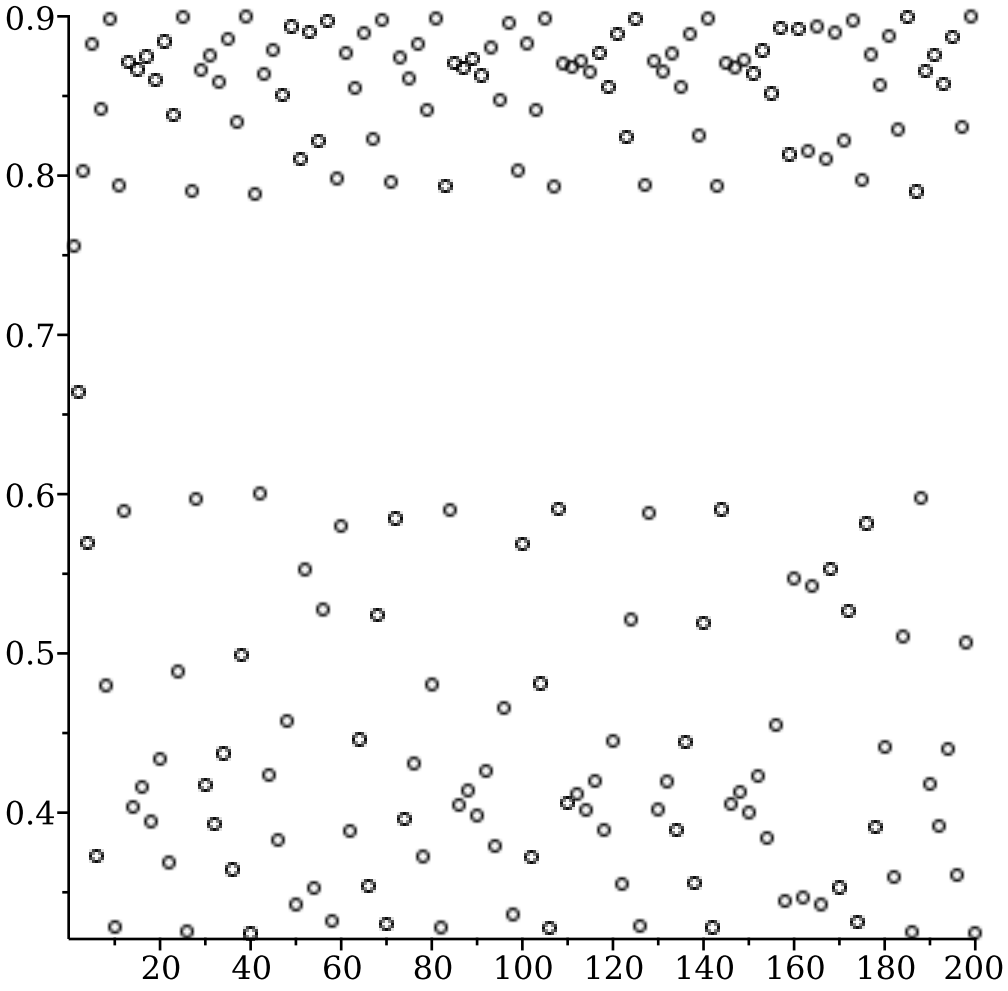
```
> for i from 1 to 200 do x := f6(x) : psi(i) := x;od:
> points := [[n, psi(n)]$n = 1..200] : with(plots) : pointplot(points, symbol
      = circle)
```




```
> x := 0.7
```

$$x := 0.7 \qquad (38)$$

```
> for i from 1 to 200 do x := f6(x) : psi(i) := x;od:
> points := [[n, psi(n)]$n = 1..200] : with(plots) : pointplot(points, symbol
      = circle)
```



> $x := 'x'$

$$x := x \tag{39}$$

> $f7 := x \to 3.8 \cdot x \cdot (1 - x)$

$$f7 := x \mapsto 3.8 \cdot x \cdot (1 - x) \tag{40}$$

> $solve(f7(x) = x, x)$

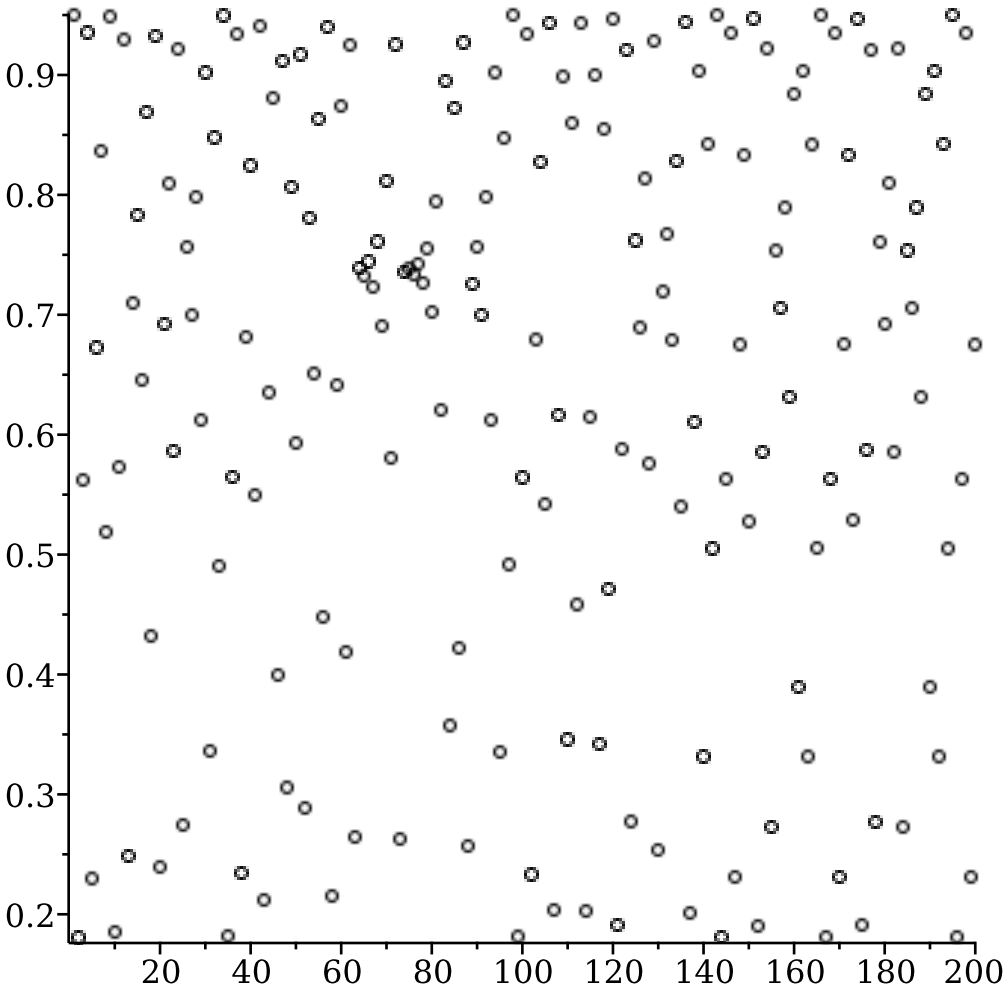
$$0., 0.7368421053 \tag{41}$$

> $x := 0.5$

$$x := 0.5 \tag{42}$$

> **for** i **from** 1 **to** 200 **do** $x := f7(x) : \mathrm{psi}(i) := x$;**od**:

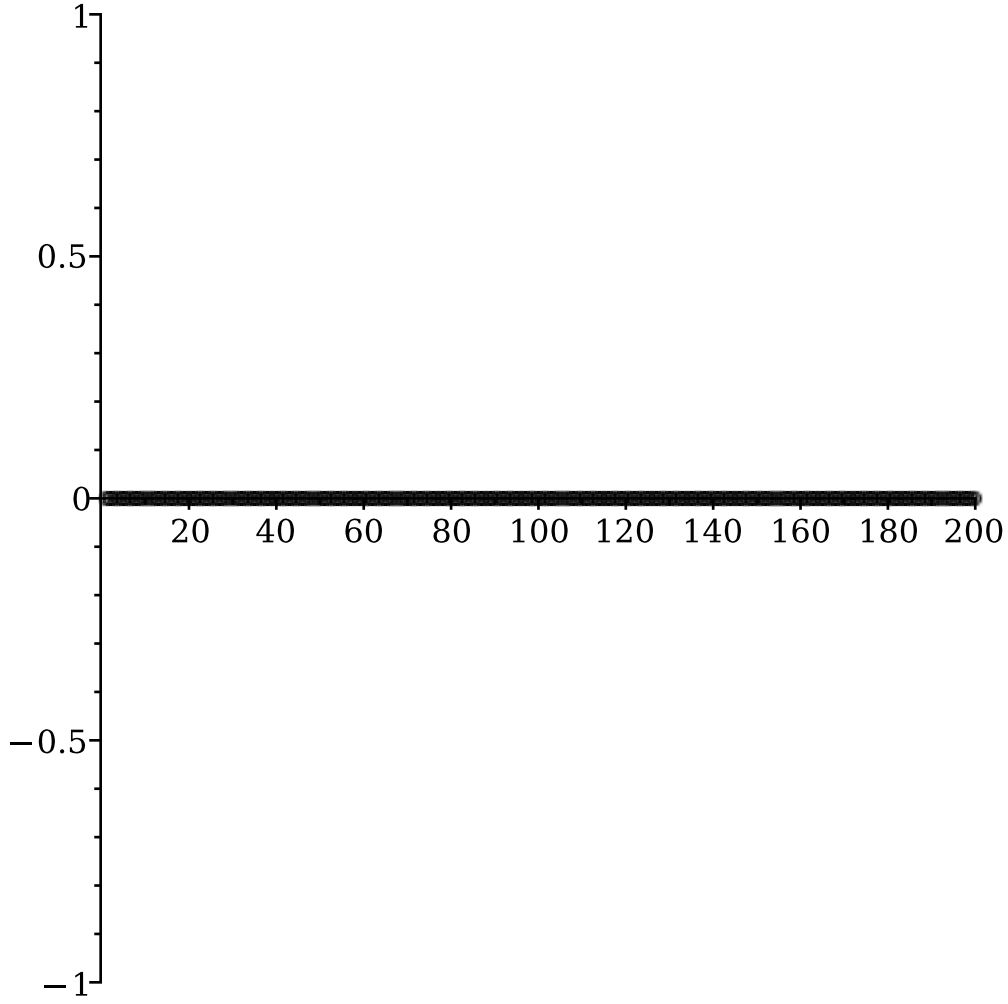
> $points := [[n, \mathrm{psi}(n)]\$n = 1..200] : with(plots) : pointplot(points, symbol = circle)$




```
> x := 0
```

$$x := 0 \qquad (43)$$

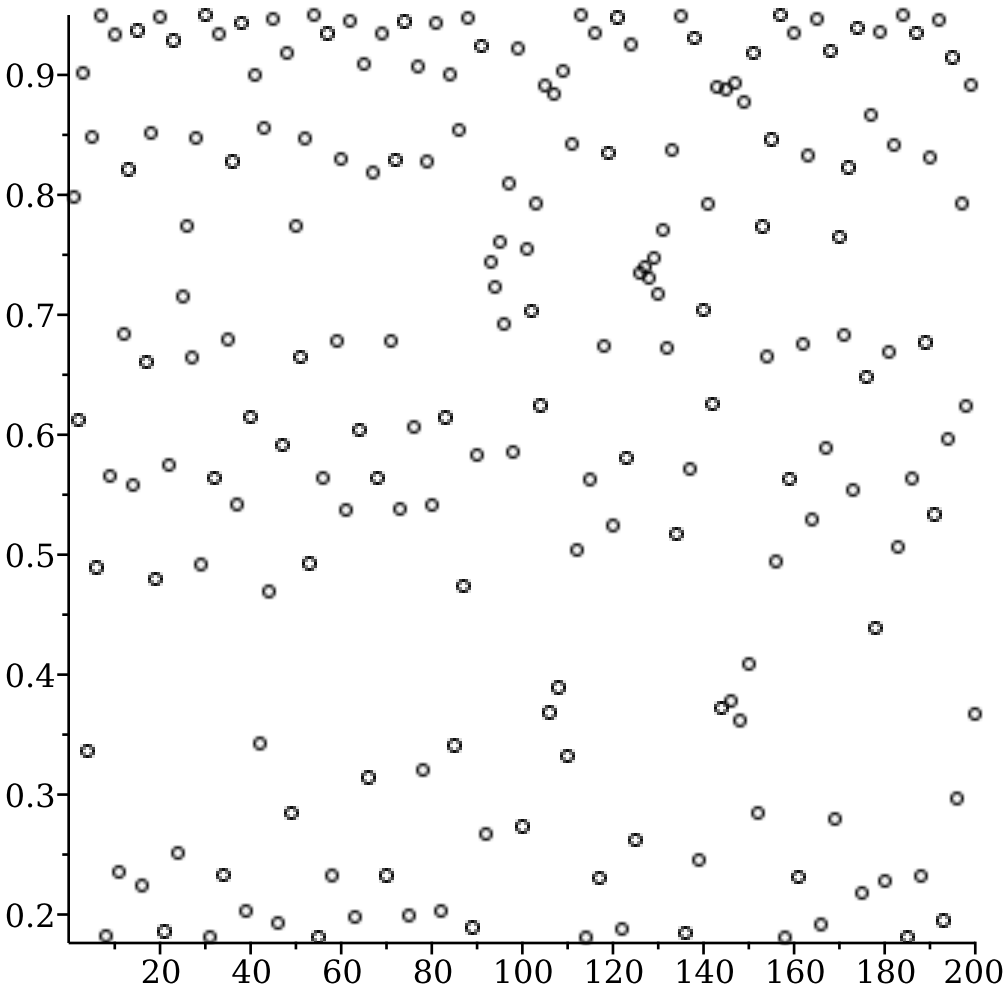
```
> for i from 1 to 200 do x := f7(x) : psi(i) := x;od:
> points := [[n, psi(n)]$n = 1..200] : with(plots) : pointplot(points, symbol
      = circle)
```




```
> x := 0.7
```

$$x := 0.7 \qquad (44)$$

```
> for i from 1 to 200 do x := f7(x) : psi(i) := x;od:
> points := [[n, psi(n)]$n = 1..200] : with(plots) : pointplot(points, symbol
      = circle)
```



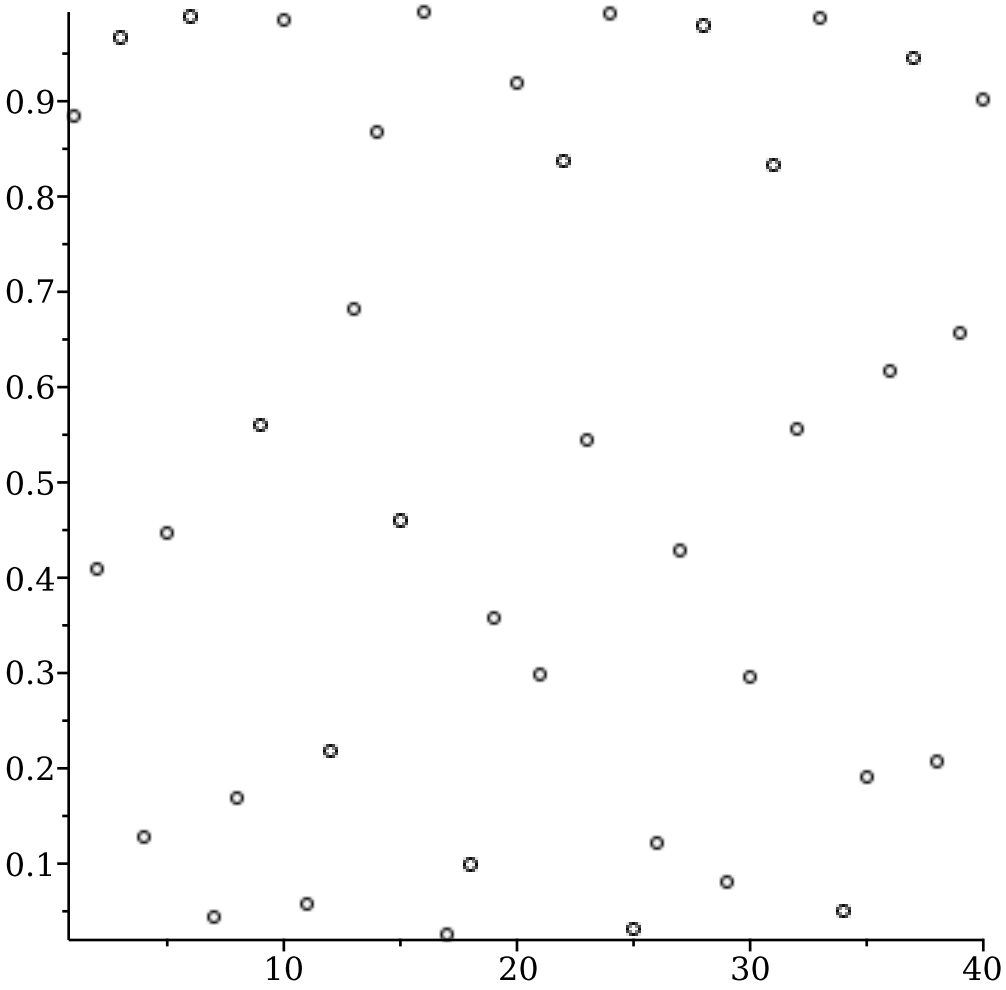

```
> fex3 := x→4·x·(1 − x)
```

$$fex3 := x \mapsto 4 \cdot x \cdot (1 - x) \tag{45}$$

```
> x := 0.67
```

$$x := 0.67 \tag{46}$$

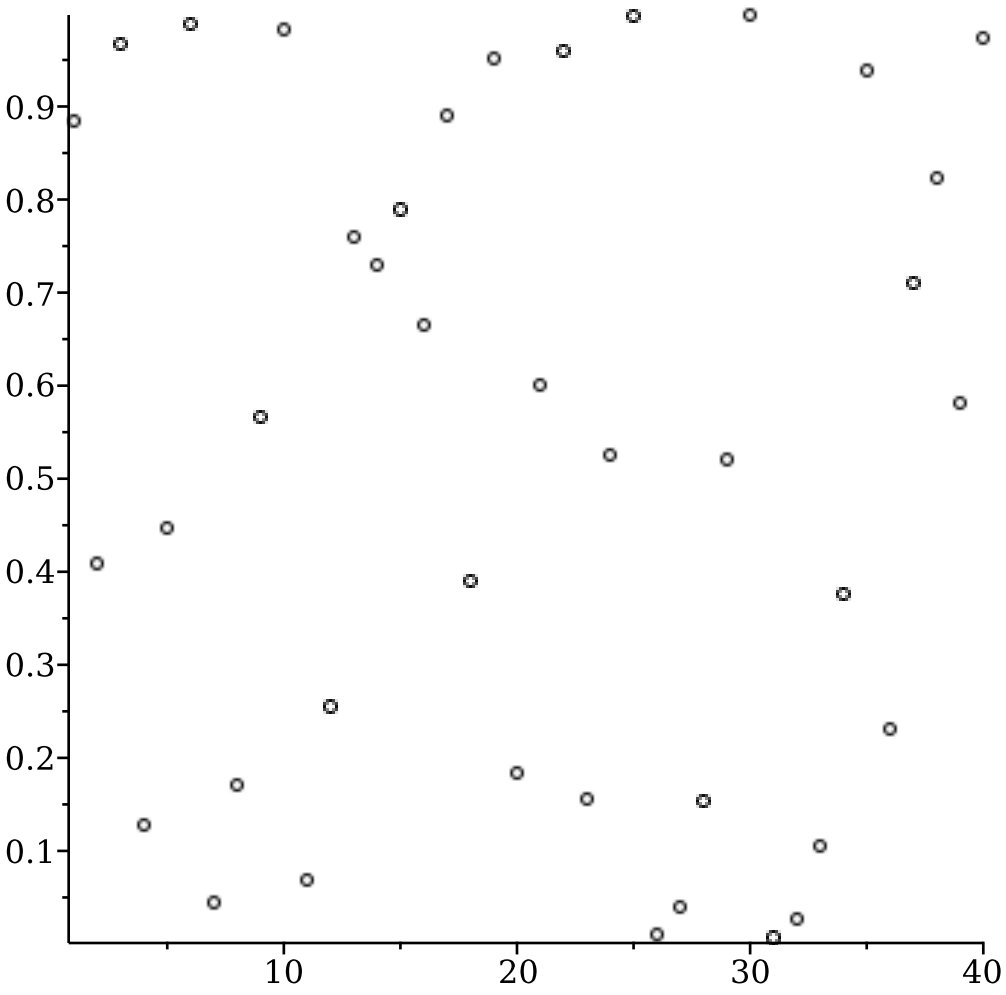
```
> for i from 1 to 40 do x := fex3(x) : psi(i) := x;od:
> points := [[n, psi(n)]$n = 1..40] : with(plots) : pointplot(points, symbol
      = circle)
```




```
> x := 0.67001
```

$$x := 0.67001 \qquad (47)$$

```
> for i from 1 to 40 do x := fex3(x) : psi(i) := x;od:
> points := [[n, psi(n)]$n = 1..40] : with(plots) : pointplot(points, symbol
      = circle)
```




```
> x := 'x'
```

$$x := x \tag{48}$$

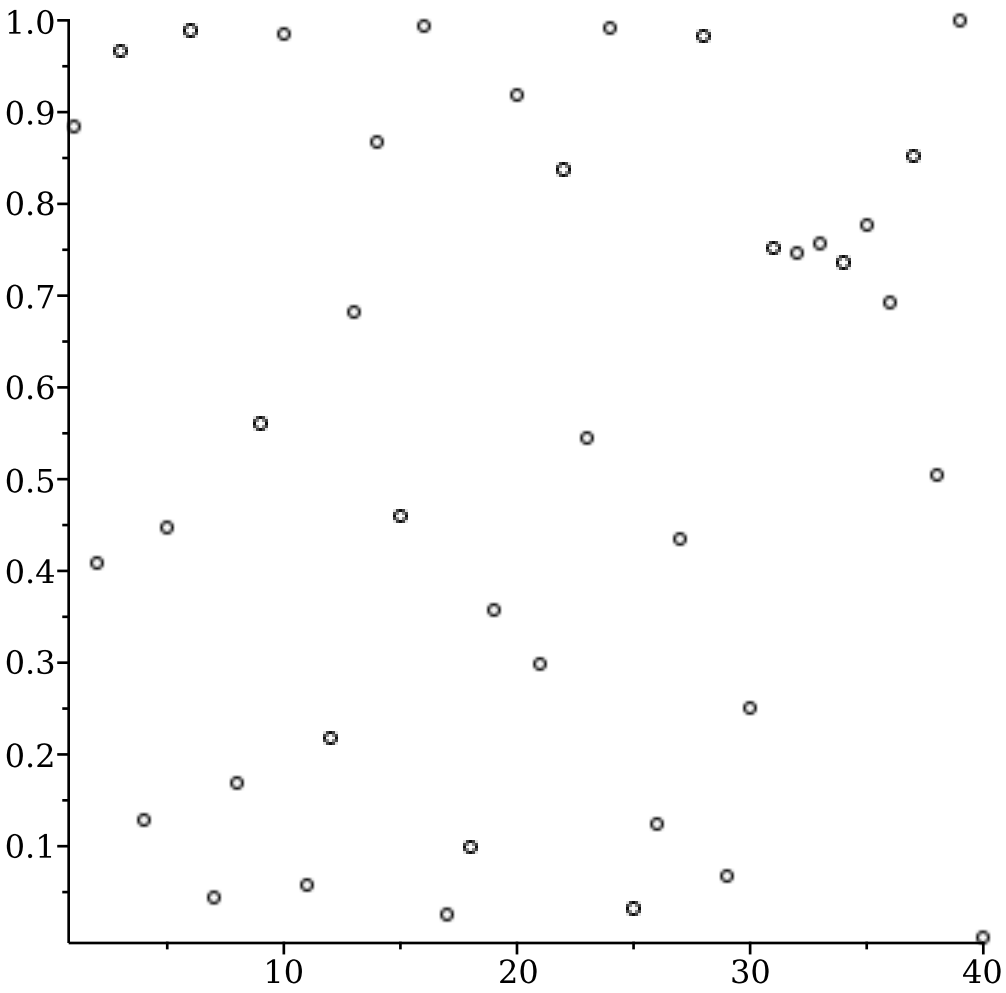
```
> fex31 := x→4·x − 4·x^2
```

$$fex31 := x \mapsto 4 \cdot x - 4 \cdot x^2 \tag{49}$$

```
> x := 0.67
```

$$x := 0.67 \tag{50}$$

```
> for i from 1 to 40 do x := fex31(x) : psi(i) := x;od:
> points := [[n, psi(n)]$n = 1..40] : with(plots) : pointplot(points, symbol
      = circle)
```




```
> restart:
> with(DEtools)
```

[*AreSimilar*, *Closure*, *DEnormal*, *DEplot*, *DEplot3d*, *DEplot_polygon*, *DFactor*, *DFactorLCLM*, *DFactorsols*, *Dchangevar*, *Desingularize*, *FindODE*, *FunctionDecomposition*, *GCRD*, *Gosper*, *Heunsols*, *Homomorphisms*, *IVPsol*, *IsHyperexponential*, *LCLM*, *MeijerGsols*, *MultiplicativeDecomposition*, *ODEInvariants*, *PDEchangecoords*, *PolynomialNormalForm*, *RationalCanonicalForm*, *ReduceHyperexp*, *RiemannPsols*, *Xchange*, *Xcommutator*, *Xgauge*, *Zeilberger*, *abelsol*, *adjoint*, *autonomous*, *bernoullisol*, *buildsol*, *buildsym*, *canoni*, *caseplot*, *casesplit*, *checkrank*, *chinisol*, *clairautsol*, *constcoeffsols*, *convertAlg*, *convertsys*, *dalembertsol*, *dcoeffs*, *de2diffop*, *dfieldplot*, *diff_table*, *diffop2de*, *dperiodic_sols*, *dpolyform*, *dsubs*, *eigenring*, *endomorphism_charpoly*, *equinv*, *eta_k*, *eulersols*, *exactsol*, *expsols*, *exterior_power*, *firint*, *firtest*, *formal_sol*, *gen_exp*, *generate_ic*, *genhomosol*, *gensys*, *hamilton_eqs*, *hypergeometricsols*, *hypergeomsols*, *hyperode*, *indicialeq*, *infgen*, *initialdata*, *integrate_sols*, *intfactor*, **(51)**

invariants, kovacicsols, leftdivision, liesol, line_int, linearsol, matrixDE, matrix_riccati, maxdimsystems, moser_reduce, muchange, mult, mutest, newton_polygon, normalG2, ode_int_y, ode_y1, odeadvisor, odepde, parametricsol, particularsol, phaseportrait, poincare, polysols, power_equivalent, rational_equivalent, ratsols, redode, reduceOrder, reduce_order, regular_parts, regularsp, remove_RootOf, riccati_system, riccatisol, rifread, rifsimp, rightdivision, rtaylor, separablesol, singularities, solve_group, super_reduce, symgen, symmetric_power, symmetric_product, symtest, transinv, translate, untranslate, varparam, zoom]

> $f := (x, y) \to 2 \cdot x \cdot y$

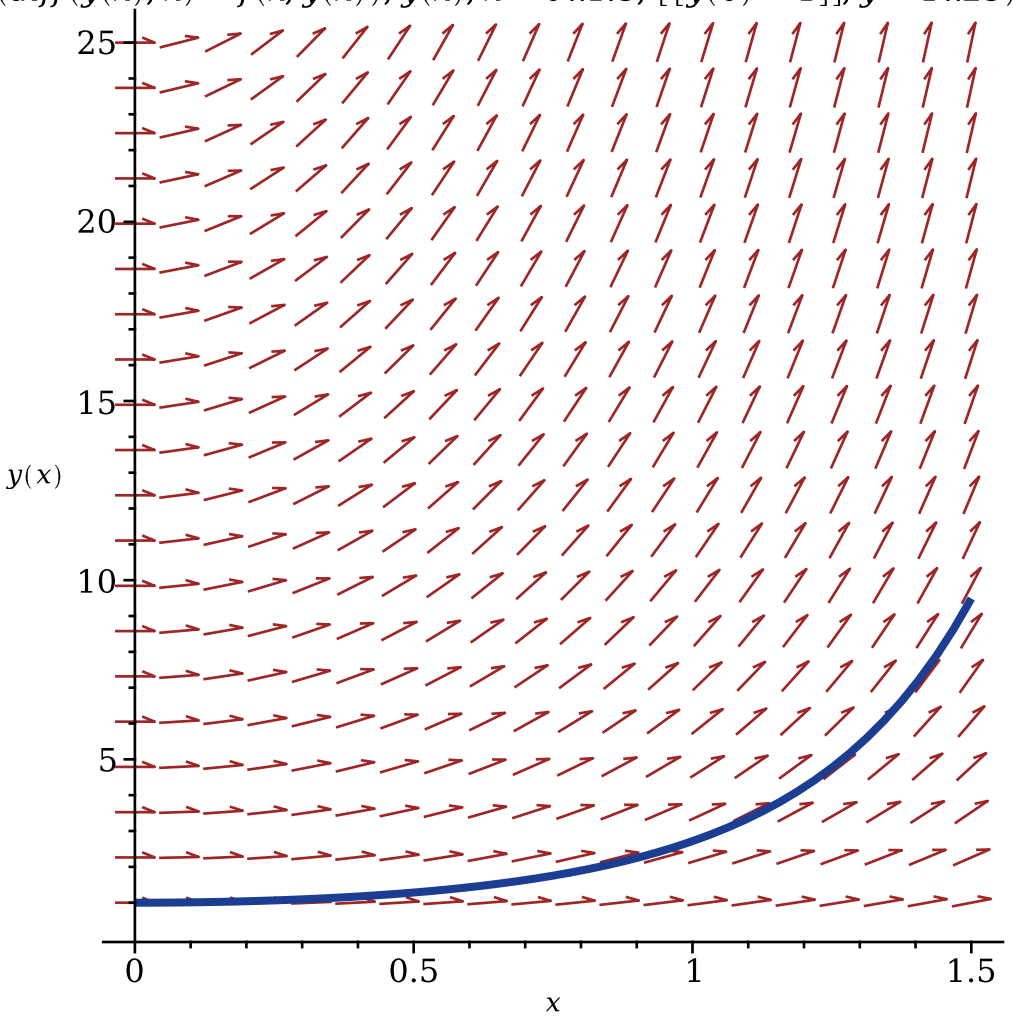
$$f := (x, y) \mapsto 2 \cdot y \cdot x \tag{52}$$

> $dsolve(\{diff(y(x), x) = f(x, y(x)), y(0) = 1\}, y(x)); \text{phi} := unapply(rhs(\%), x)$

$$y(x) = e^{x^2}$$

$$\phi := x \mapsto e^{x^2} \tag{53}$$

> $DEplot(diff(y(x), x) = f(x, y(x)), y(x), x = 0..1.5, [[y(0) = 1]], y = 1..25)$



```
> h := 0.1
```

$$h := 0.1 \tag{54}$$

```
> x := 0; y := 1
```

$$x := 0$$

$$y := 1 \tag{55}$$

```
> for i from 1 to 15 do y := y + h·f(x, y) : psi(i) := y : x := x + h : print(x, y,
    phi(x), abs(y − phi(x))); od
```

$$y := 273.7075287$$

$$\psi(1) := 273.7075287$$

$$x := 2.6$$

$$2.6,\ 273.7075287,\ 862.6421958,\ 588.9346671$$

$$y := 416.0354436$$

$$\psi(2) := 416.0354436$$

$$x := 2.7$$

$$2.7,\ 416.0354436,\ 1465.570697,\ 1049.535253$$

$$y := 640.6945832$$

$$\psi(3) := 640.6945832$$

$$x := 2.8$$

$$2.8,\ 640.6945832,\ 2540.204834,\ 1899.510251$$

$$y := 999.4835498$$

$$\psi(4) := 999.4835498$$

$$x := 2.9$$

$$2.9,\ 999.4835498,\ 4491.760512,\ 3492.276962$$

$$y := 1579.184009$$

$$\psi(5) := 1579.184009$$

$$x := 3.0$$

$$3.0,\ 1579.184009,\ 8103.083928,\ 6523.899919$$

$$y := 2526.694414$$

$$\psi(6) := 2526.694414$$

$$x := 3.1$$

$$3.1,\ 2526.694414,\ 14913.17009,\ 12386.47568$$

$$y := 4093.244951$$

$$\psi(7) := 4093.244951$$

$$x := 3.2$$

$$3.2,\ 4093.244951,\ 28001.12593,\ 23907.88098$$

$$y := 6712.921719$$

$$\psi(8) := 6712.921719$$

$$x := 3.3$$

$$3.3,\ 6712.921719,\ 53637.29997,\ 46924.37825$$

$$y := 11143.45005$$

$$\psi(9) := 11143.45005$$

$$x := 3.4$$

$$3.4,\ 11143.45005,\ 104820.0134,\ 93676.56335$$

$$y := 18720.99608$$

$$\psi(10) := 18720.99608$$

$$x := 3.5$$

$$3.5,\ 18720.99608,\ 208981.2889,\ 190260.2928$$

$$y := 31825.69334$$

$$\psi(11) := 31825.69334$$

$$x := 3.6$$

$$3.6,\ 31825.69334,\ 425066.1148,\ 393240.4215$$

$$y := 54740.19254$$

$$\psi(12) := 54740.19254$$

$$x := 3.7$$

$$3.7,\ 54740.19254,\ 882046.4518,\ 827306.2593$$

$$y := 95247.93502$$

$$\psi(13) := 95247.93502$$

$$x := 3.8$$

$$3.8,\ 95247.93502,\ 1.867292353 \times 10^{6},\ 1.772044418 \times 10^{6}$$

$$y := 167636.3656$$

$$\psi(14) := 167636.3656$$

$$x := 3.9$$

$$3.9,\ 167636.3656,\ 4.032915010 \times 10^{6},\ 3.865278644 \times 10^{6}$$

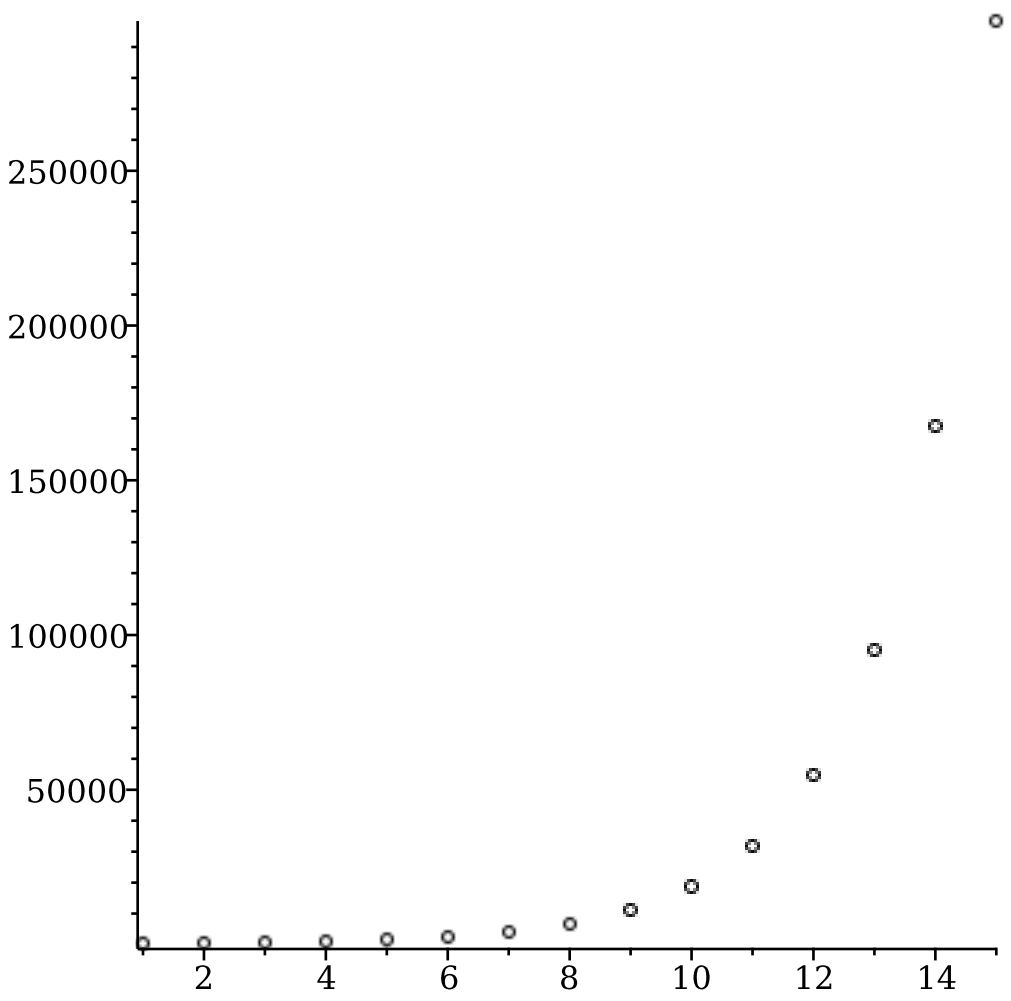
$$y := 298392.7308$$

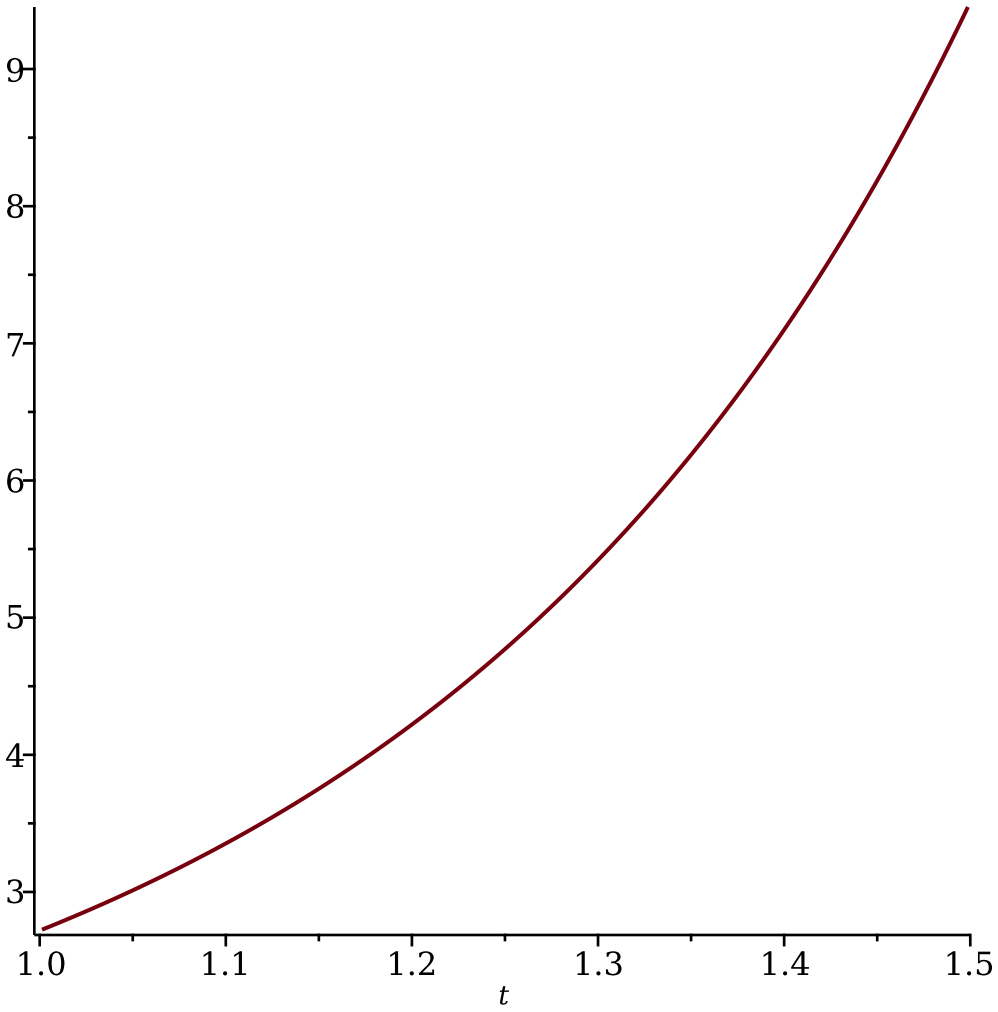
$$\psi(15) := 298392.7308$$

$$x := 4.0$$

$$4.0,\ 298392.7308,\ 8.886110521 \times 10^{6},\ 8.587717790 \times 10^{6} \qquad (56)$$

```
> points := [[n, psi(n)]$n = 1..15]: with(plots): pointplot(points, symbol
      = circle); plot(phi(t), t = 1..1.5)
```


250000
200000
150000
100000
50000
2
4
6
8
10
12
14




```
> restart
> phi := x→exp(x^2)
```

$$\phi := x \mapsto e^{x^2} \quad (57)$$

```
> h := 0.1
```

$$h := 0.1 \quad (58)$$

```
> x := 0; y := 1
```

$$x := 0$$

$$y := 1 \quad (59)$$

```
> f := (x, y)→2·x·y
```

$$f := (x, y) \mapsto 2 \cdot y \cdot x \quad (60)$$

```
> for i from 1 to 10 do y := y + h/2·f(x, y) + h/2·f(x + h, y + h·f(x, y)) : psi(i) := y :
    x := x + h : print(x, y, phi(x), abs(y − phi(x))); od
```

$$y := 12.69561889$$

$$\psi(1) := 12.69561889$$

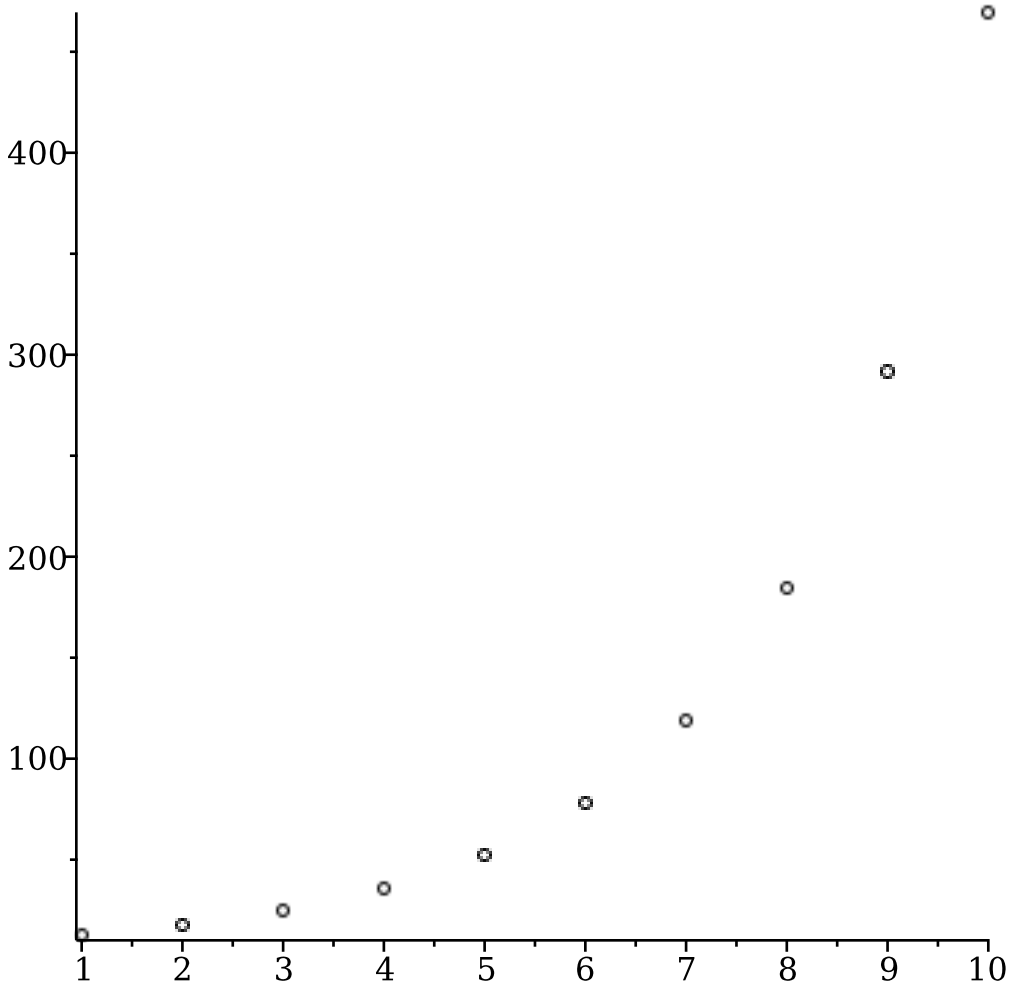
$$x := 1.6$$

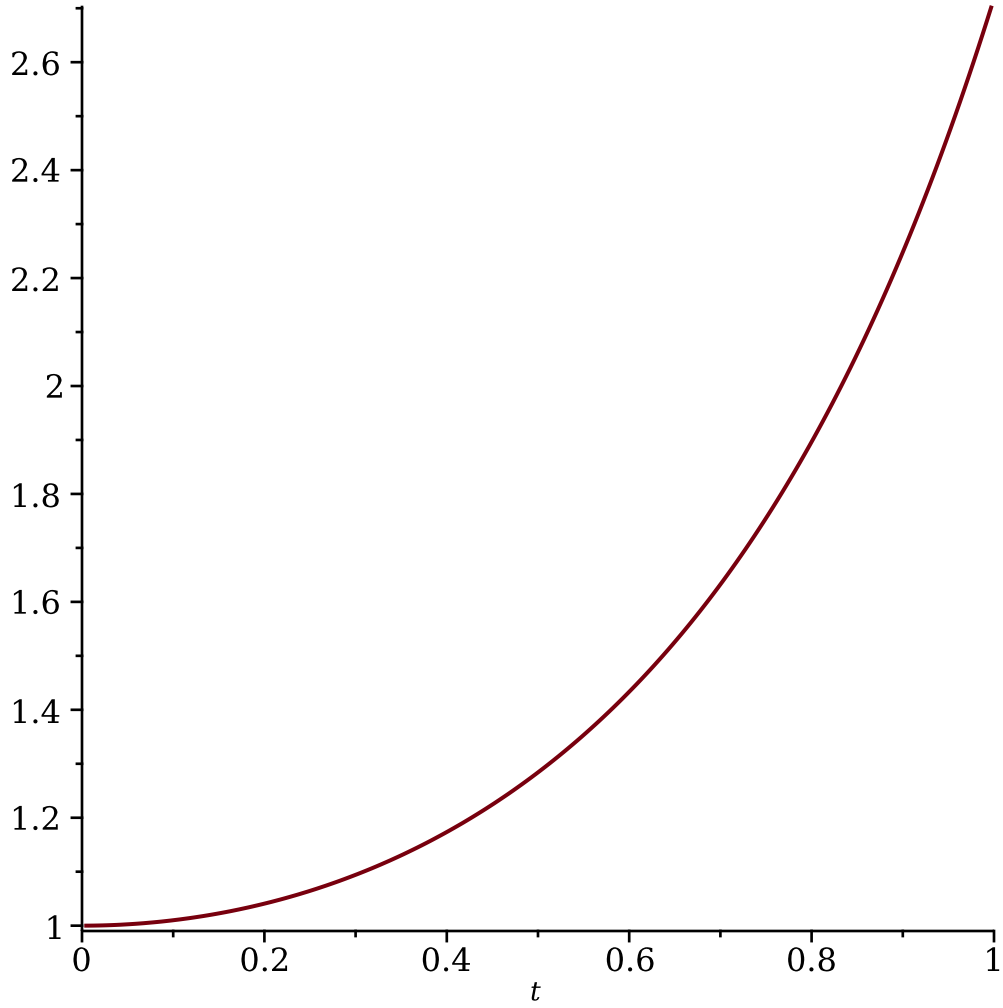
$$1.6,\ 12.69561889,\ 12.93581732,\ 0.24019843$$

$$y := 17.57581479$$
$$\psi(2) := 17.57581479$$
$$x := 1.7$$
$$1.7, 17.57581479, 17.99330960, 0.41749481$$
$$y := 24.80298983$$
$$\psi(3) := 24.80298983$$
$$x := 1.8$$
$$1.8, 24.80298983, 25.53372175, 0.73073192$$
$$y := 35.67662057$$
$$\psi(4) := 35.67662057$$
$$x := 1.9$$
$$1.9, 35.67662057, 36.96605281, 1.28943224$$
$$y := 52.30192576$$
$$\psi(5) := 52.30192576$$
$$x := 2.0$$
$$2.0, 52.30192576, 54.59815003, 2.29622427$$
$$y := 78.13907708$$
$$\psi(6) := 78.13907708$$
$$x := 2.1$$
$$2.1, 78.13907708, 82.26946350, 4.13038642$$
$$y := 118.9589310$$
$$\psi(7) := 118.9589310$$
$$x := 2.2$$
$$2.2, 118.9589310, 126.4693517, 7.5104207$$
$$y := 184.5290937$$
$$\psi(8) := 184.5290937$$
$$x := 2.3$$
$$2.3, 184.5290937, 198.3434254, 13.8143317$$
$$y := 291.6297796$$
$$\psi(9) := 291.6297796$$
$$x := 2.4$$
$$2.4, 291.6297796, 317.3483289, 25.7185493$$
$$y := 469.5239451$$
$$\psi(10) := 469.5239451$$
$$x := 2.5$$
$$2.5, 469.5239451, 518.0128247, 48.4888796 \quad (61)$$

```
> points := [[n, psi(n)]$n = 1..10]: with(plots): pointplot(points, symbol
```

$= circle); plot(\mathrm{phi}(t), t = 0..1)$






```
> psi(0.5)
```

$$\psi(0.5) \tag{62}$$

```
> restart
> with(DEtools)
```

[*AreSimilar*, *Closure*, *DEnormal*, *DEplot*, *DEplot3d*, *DEplot_polygon*, *DFactor*, *DFactorLCLM*, *DFactorsols*, *Dchangevar*, *Desingularize*, *FindODE*, *FunctionDecomposition*, *GCRD*, *Gosper*, *Heunsols*, *Homomorphisms*, *IVPsol*, *IsHyperexponential*, *LCLM*, *MeijerGsols*, *MultiplicativeDecomposition*, *ODEInvariants*, *PDEchangecoords*, *PolynomialNormalForm*, *RationalCanonicalForm*, *ReduceHyperexp*, *RiemannPsols*, *Xchange*, *Xcommutator*, *Xgauge*, *Zeilberger*, *abelsol*, *adjoint*, *autonomous*, *bernoullisol*, *buildsol*, *buildsym*, *canoni*, *caseplot*, *casesplit*, *checkrank*, *chinisol*, *clairautsol*, *constcoeffsols*, *convertAlg*, *convertsys*, *dalembertsol*, *dcoeffs*, *de2diffop*, *dfieldplot*, *diff_table*, *diffop2de*, *dperiodic_sols*, *dpolyform*, *dsubs*, *eigenring*, *endomorphism_charpoly*, *equinv*, *eta_k*, *eulersols*, *exactsol*, *expsols*, *exterior_power*, *firint*, *firtest*, *formal_sol*, *gen_exp*, *generate_ic*, (63)

genhomosol, gensys, hamilton_eqs, hypergeometricsols, hypergeomsols, hyperode, indicialeq, infgen, initialdata, integrate_sols, intfactor, invariants, kovacicsols, leftdivision, liesol, line_int, linearsol, matrixDE, matrix_riccati, maxdimsystems, moser_reduce, muchange, mult, mutest, newton_polygon, normalG2, ode_int_y, ode_y1, odeadvisor, odepde, parametricsol, particularsol, phaseportrait, poincare, polysols, power_equivalent, rational_equivalent, ratsols, redode, reduceOrder, reduce_order, regular_parts, regularsp, remove_RootOf, riccati_system, riccatisol, rifread, rifsimp, rightdivision, rtaylor, separablesol, singularities, solve_group, super_reduce, symgen, symmetric_power, symmetric_product, symtest, transinv, translate, untranslate, varparam, zoom]

> $f := (x, y) \to y^2 + x^2$

$$f := (x, y) \mapsto y^2 + x^2 \tag{64}$$

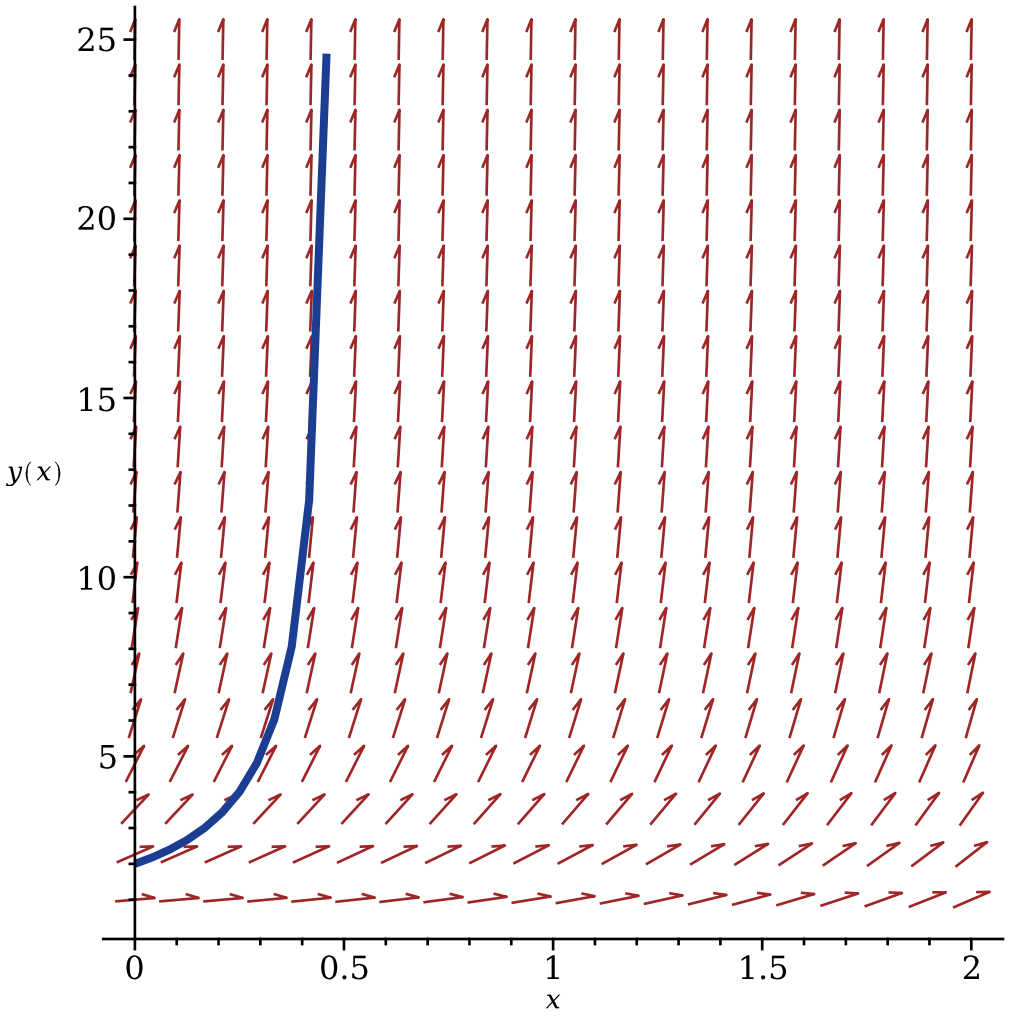
> *dsolve*({*diff*($y(x)$, x) = $f(x, y(x))$, $y(0) = 0$}); phi := *unapply*(*rhs*(%), x)

$$y(x) = -\frac{\left(-\mathrm{BesselJ}\left(-\frac{3}{4}, \frac{x^2}{2}\right) + \mathrm{BesselY}\left(-\frac{3}{4}, \frac{x^2}{2}\right)\right) x}{-\mathrm{BesselJ}\left(\frac{1}{4}, \frac{x^2}{2}\right) + \mathrm{BesselY}\left(\frac{1}{4}, \frac{x^2}{2}\right)}$$

$$\phi := x \mapsto -\frac{\left(-\mathrm{BesselJ}\left(-\frac{3}{4}, \frac{x^2}{2}\right) + \mathrm{BesselY}\left(-\frac{3}{4}, \frac{x^2}{2}\right)\right) \cdot x}{-\mathrm{BesselJ}\left(\frac{1}{4}, \frac{x^2}{2}\right) + \mathrm{BesselY}\left(\frac{1}{4}, \frac{x^2}{2}\right)} \tag{65}$$

> *DEplot*(*diff*($y(x)$, x) = $f(x, y(x))$, $y(x)$, $x = 0..2$, [[$y(0) = 2$]], $y = 1..25$)

Warning, plot may be incomplete, the following errors(s) were issued:
cannot evaluate the solution further right of .49896524, probably a singularity



> $x := 0; y := 0; h := 0.1$

$$x := 0$$
$$y := 0$$
$$h := 0.1 \quad \textbf{(66)}$$

> **for** i **from** 1 **to** 20 **do** $y := y + h \cdot f(x, y)$; psi$(i) := y : x := x + h : print(x, y,$ phi$(x),$ abs$(y -$ phi$(x)))$;**od**

$$y := 0.$$
$$\psi(1) := 0.$$
$$x := 0.1$$
$$0.1, 0., 0.0003333349060, 0.0003333349060$$
$$y := 0.001$$
$$\psi(2) := 0.001$$
$$x := 0.2$$
$$0.2, 0.001, 0.002666869814, 0.001666869814$$
$$y := 0.0050001$$
$$\psi(3) := 0.0050001$$
$$x := 0.3$$

0.3, 0.0050001, 0.009003473189, 0.004003373189

$y := 0.01400260010$

$\psi(4) := 0.01400260010$

$x := 0.4$

0.4, 0.01400260010, 0.02135938017, 0.00735678007

$y := 0.03002220738$

$\psi(5) := 0.03002220738$

$x := 0.5$

0.5, 0.03002220738, 0.04179114620, 0.01176893882

$y := 0.05511234067$

$\psi(6) := 0.05511234067$

$x := 0.6$

0.6, 0.05511234067, 0.07244786117, 0.01733552050

$y := 0.09141607768$

$\psi(7) := 0.09141607768$

$x := 0.7$

0.7, 0.09141607768, 0.1156598536, 0.02424377592

$y := 0.1412517676$

$\psi(8) := 0.1412517676$

$x := 0.8$

0.8, 0.1412517676, 0.1740802646, 0.0328284970

$y := 0.2072469738$

$\psi(9) := 0.2072469738$

$x := 0.9$

0.9, 0.2072469738, 0.2509066825, 0.0436597087

$y := 0.2925421046$

$\psi(10) := 0.2925421046$

$x := 1.0$

1.0, 0.2925421046, 0.3502318440, 0.0576897394

$y := 0.4011001929$

$\psi(11) := 0.4011001929$

$x := 1.1$

1.1, 0.4011001929, 0.4776170219, 0.0765168290

$y := 0.5381883294$

$\psi(12) := 0.5381883294$

$x := 1.2$

1.2, 0.5381883294, 0.6410767262, 0.1028883968

$$y := 0.7111529972$$

$$\psi(13) := 0.7111529972$$

$$x := 1.3$$

$$1.3, 0.7111529972, 0.8528799930, 0.1417269958$$

$$y := 0.9307268557$$

$$\psi(14) := 0.9307268557$$

$$x := 1.4$$

$$1.4, 0.9307268557, 1.133112675, 0.2023858193$$

$$y := 1.213352104$$

$$\psi(15) := 1.213352104$$

$$x := 1.5$$

$$1.5, 1.213352104, 1.517447543, 0.304095439$$

$$y := 1.585574437$$

$$\psi(16) := 1.585574437$$

$$x := 1.6$$

$$1.6, 1.585574437, 2.076423381, 0.490848944$$

$$y := 2.092979066$$

$$\psi(17) := 2.092979066$$

$$x := 1.7$$

$$1.7, 2.092979066, 2.972797219, 0.879818153$$

$$y := 2.820035203$$

$$\psi(18) := 2.820035203$$

$$x := 1.8$$

$$1.8, 2.820035203, 4.688130710, 1.868095507$$

$$y := 3.939295058$$

$$\psi(19) := 3.939295058$$

$$x := 1.9$$

$$1.9, 3.939295058, 9.566995342, 5.627700284$$

$$y := 5.852099613$$

$$\psi(20) := 5.852099613$$

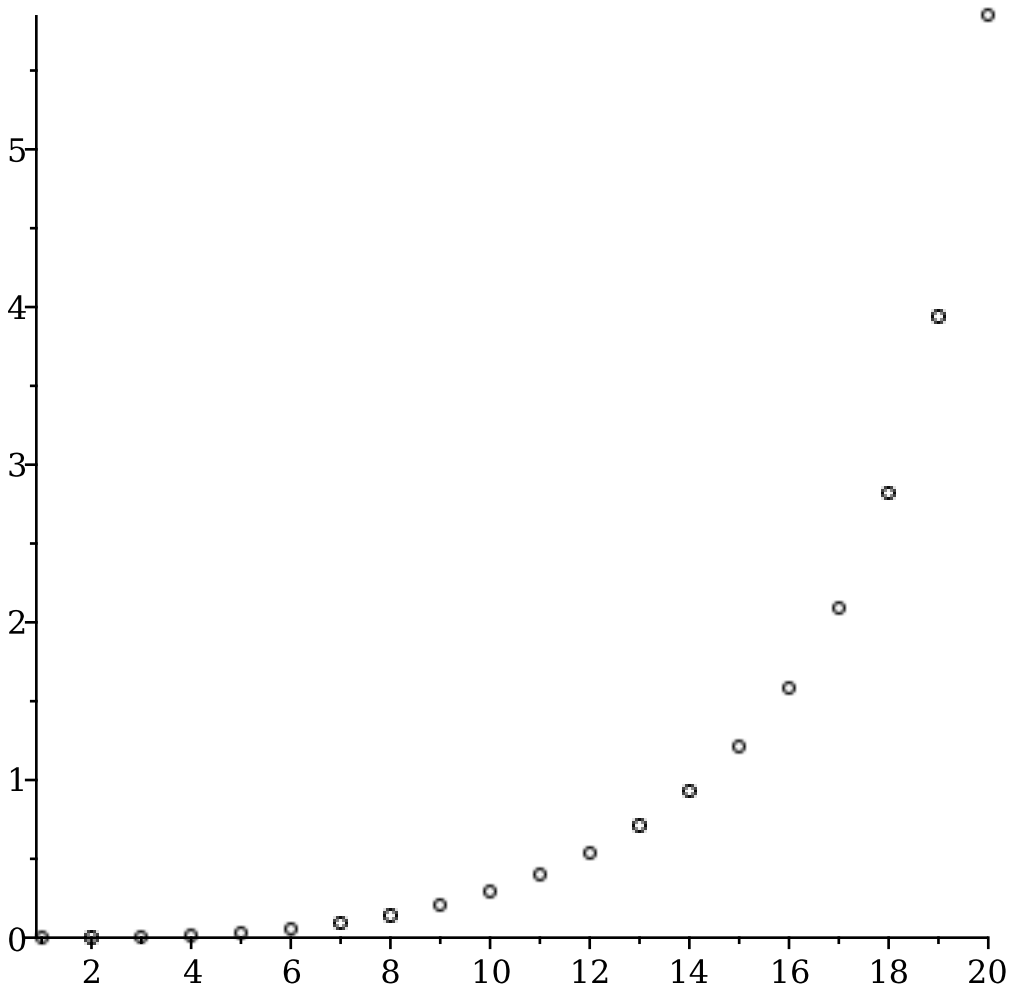
$$x := 2.0$$

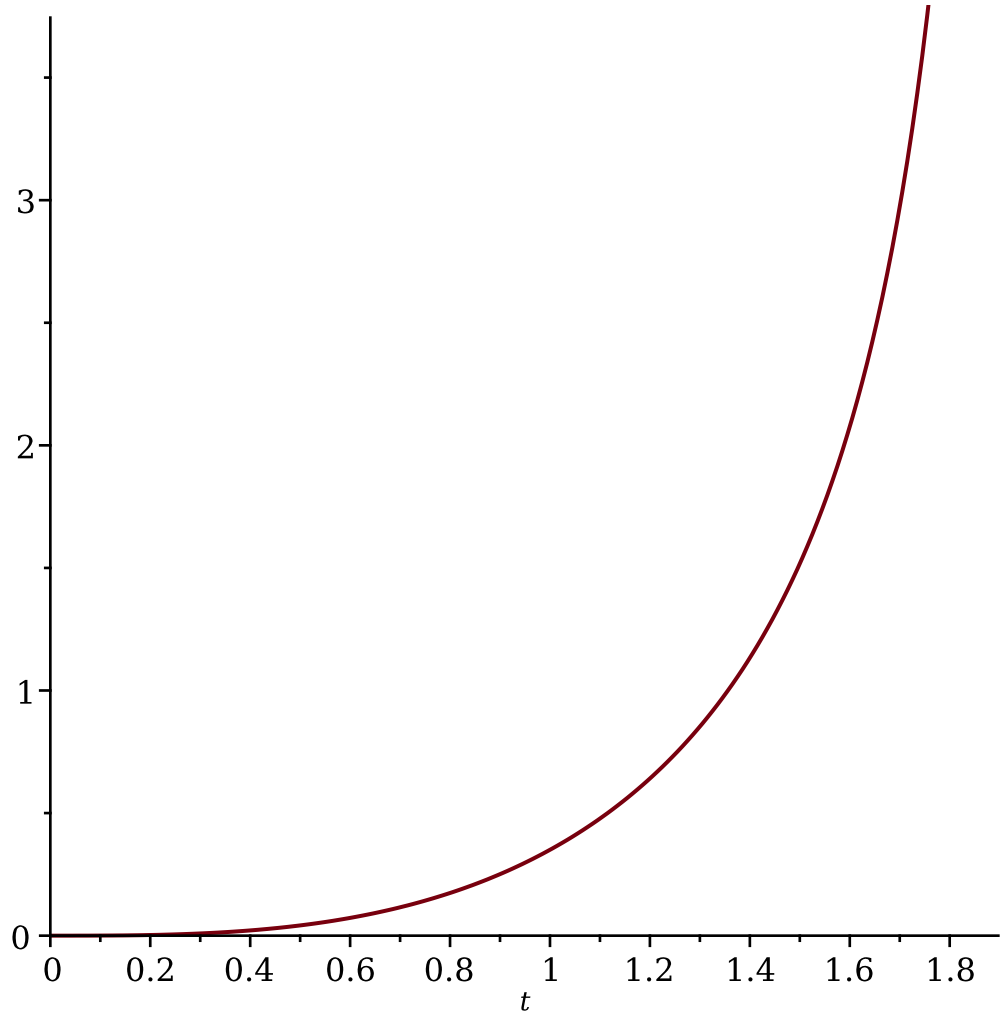
$$2.0, 5.852099613, 317.7224621, 311.8703625 \quad (67)$$

```
> y :='y'
```

$$y := y \quad (68)$$

```
> points := [[n, psi(n)]$n = 1..20]: with(plots): pointplot(points, symbol
      = circle); plot(phi(t), t = 0..2)
```


0
1
2
3
4
5
2
4
6
8
10
12
14
16
18
20




```
> restart
> with(DEtools)
```

[*AreSimilar*, *Closure*, *DEnormal*, *DEplot*, *DEplot3d*, *DEplot_polygon*, *DFactor*, *DFactorLCLM*, *DFactorsols*, *Dchangevar*, *Desingularize*, *FindODE*, *FunctionDecomposition*, *GCRD*, *Gosper*, *Heunsols*, *Homomorphisms*, *IVPsol*, *IsHyperexponential*, *LCLM*, *MeijerGsols*, *MultiplicativeDecomposition*, *ODEInvariants*, *PDEchangecoords*, *PolynomialNormalForm*, *RationalCanonicalForm*, *ReduceHyperexp*, *RiemannPsols*, *Xchange*, *Xcommutator*, *Xgauge*, *Zeilberger*, *abelsol*, *adjoint*, *autonomous*, *bernoullisol*, *buildsol*, *buildsym*, *canoni*, *caseplot*, *casesplit*, *checkrank*, *chinisol*, *clairautsol*, *constcoeffsols*, *convertAlg*, *convertsys*, *dalembertsol*, *dcoeffs*, *de2diffop*, *dfieldplot*, *diff_table*, *diffop2de*, *dperiodic_sols*, *dpolyform*, *dsubs*, *eigenring*, *endomorphism_charpoly*, *equinv*, *eta_k*, *eulersols*, *exactsol*, *expsols*, *exterior_power*, *firint*, *firtest*, *formal_sol*, *gen_exp*, *generate_ic*, *genhomosol*, *gensys*, *hamilton_eqs*, *hypergeometricsols*, *hypergeomsols*, *hyperode*, *indicialeq*, *infgen*, *initialdata*, *integrate_sols*, *intfactor*, **(69)**

invariants, kovacicsols, leftdivision, liesol, line_int, linearsol, matrixDE, matrix_riccati, maxdimsystems, moser_reduce, muchange, mult, mutest, newton_polygon, normalG2, ode_int_y, ode_y1, odeadvisor, odepde, parametricsol, particularsol, phaseportrait, poincare, polysols, power_equivalent, rational_equivalent, ratsols, redode, reduceOrder, reduce_order, regular_parts, regularsp, remove_RootOf, riccati_system, riccatisol, rifread, rifsimp, rightdivision, rtaylor, separablesol, singularities, solve_group, super_reduce, symgen, symmetric_power, symmetric_product, symtest, transinv, translate, untranslate, varparam, zoom]

> $f := (x, y) \to y^2 + x^2$

$$f := (x, y) \mapsto y^2 + x^2 \tag{70}$$

> *dsolve*({*diff*($y(x)$, x) = $f(x, y(x))$, $y(0) = 0$}, $y(x)$); phi := *unapply*(*rhs*(%), x)

$$y(x) = -\frac{\left(-\text{BesselJ}\left(-\frac{3}{4}, \frac{x^2}{2}\right) + \text{BesselY}\left(-\frac{3}{4}, \frac{x^2}{2}\right)\right) x}{-\text{BesselJ}\left(\frac{1}{4}, \frac{x^2}{2}\right) + \text{BesselY}\left(\frac{1}{4}, \frac{x^2}{2}\right)}$$

$$\phi := x \mapsto -\frac{\left(-\text{BesselJ}\left(-\frac{3}{4}, \frac{x^2}{2}\right) + \text{BesselY}\left(-\frac{3}{4}, \frac{x^2}{2}\right)\right) \cdot x}{-\text{BesselJ}\left(\frac{1}{4}, \frac{x^2}{2}\right) + \text{BesselY}\left(\frac{1}{4}, \frac{x^2}{2}\right)} \tag{71}$$

> $h := 0.1$; $x := 0$; $y := 0$

$$h := 0.1$$
$$x := 0$$
$$y := 0 \tag{72}$$

> **for** i **from** 1 **to** 20 **do** $y := y + \frac{h}{2} \cdot f(x, y) + \frac{h}{2} \cdot f(x + h, y + h \cdot f(x, y))$: psi(i) := y : $x := x + h$: *print*(x, y, phi(x), abs(y − phi(x))) : **od**

$$y := 0.0005000000000$$
$$\psi(1) := 0.0005000000000$$
$$x := 0.1$$
$$0.1,\ 0.0005000000000,\ 0.0003333349060,\ 0.0001666650940$$
$$y := 0.003000125004$$
$$\psi(2) := 0.003000125004$$
$$x := 0.2$$
$$0.2,\ 0.003000125004,\ 0.002666869814,\ 0.000333255190$$
$$y := 0.009503025760$$
$$\psi(3) := 0.009503025760$$

$$x := 0.3$$

$$0.3, 0.009503025760, 0.009003473189, 0.000499552571$$

$$y := 0.02202467595$$

$$\psi(4) := 0.02202467595$$

$$x := 0.4$$

$$0.4, 0.02202467595, 0.02135938017, 0.00066529578$$

$$y := 0.04262140864$$

$$\psi(5) := 0.04262140864$$

$$x := 0.5$$

$$0.5, 0.04262140864, 0.04179114620, 0.00083026244$$

$$y := 0.07344210066$$

$$\psi(6) := 0.07344210066$$

$$x := 0.6$$

$$0.6, 0.07344210066, 0.07244786117, 0.00099423949$$

$$y := 0.1168165840$$

$$\psi(7) := 0.1168165840$$

$$x := 0.7$$

$$0.7, 0.1168165840, 0.1156598536, 0.0011567304$$

$$y := 0.1753963673$$

$$\psi(8) := 0.1753963673$$

$$x := 0.8$$

$$0.8, 0.1753963673, 0.1740802646, 0.0013161027$$

$$y := 0.2523742135$$

$$\psi(9) := 0.2523742135$$

$$x := 0.9$$

$$0.9, 0.2523742135, 0.2509066825, 0.0014675310$$

$$y := 0.3518301326$$

$$\psi(10) := 0.3518301326$$

$$x := 1.0$$

$$1.0, 0.3518301326, 0.3502318440, 0.0015982886$$

$$y := 0.4792938348$$

$$\psi(11) := 0.4792938348$$

$$x := 1.1$$

$$1.1, 0.4792938348, 0.4776170219, 0.0016768129$$

$$y := 0.6427029949$$

$$\psi(12) := 0.6427029949$$

$$x := 1.2$$

1.2, 0.6427029949, 0.6410767262, 0.0016262687

$y := 0.8541363558$

$\psi(13) := 0.8541363558$

$x := 1.3$

1.3, 0.8541363558, 0.8528799930, 0.0012563628

$y := 1.133184603$

$\psi(14) := 1.133184603$

$x := 1.4$

1.4, 1.133184603, 1.133112675, 0.000071928

$y := 1.514119178$

$\psi(15) := 1.514119178$

$x := 1.5$

1.5, 1.514119178, 1.517447543, 0.003328365

$y := 2.062972003$

$\psi(16) := 2.062972003$

$x := 1.6$

1.6, 2.062972003, 2.076423381, 0.013451378

$y := 2.924894430$

$\psi(17) := 2.924894430$

$x := 1.7$

1.7, 2.924894430, 2.972797219, 0.047902789

$y := 4.487143656$

$\psi(18) := 4.487143656$

$x := 1.8$

1.8, 4.487143656, 4.688130710, 0.200987054

$y := 8.165117641$

$\psi(19) := 8.165117641$

$x := 1.9$

1.9, 8.165117641, 9.566995342, 1.401877701

$y := 23.42048639$

$\psi(20) := 23.42048639$

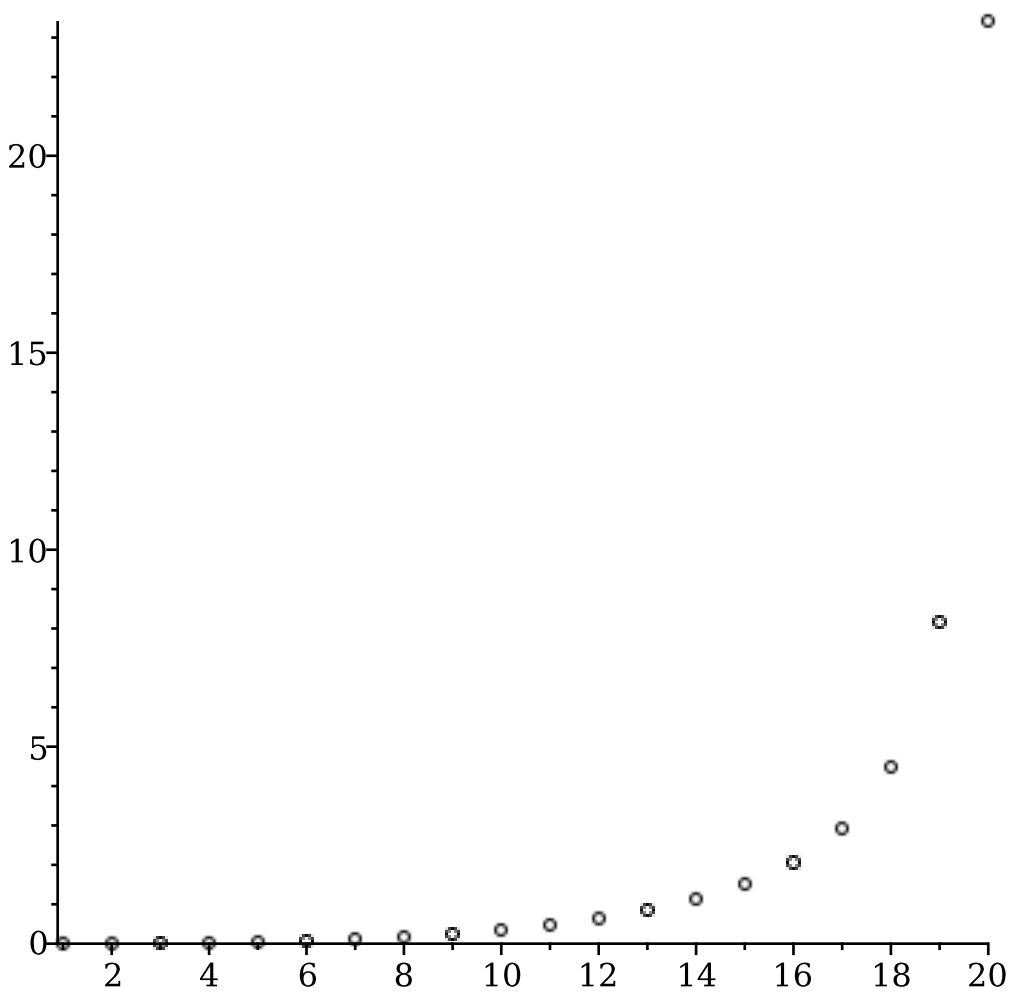
$x := 2.0$

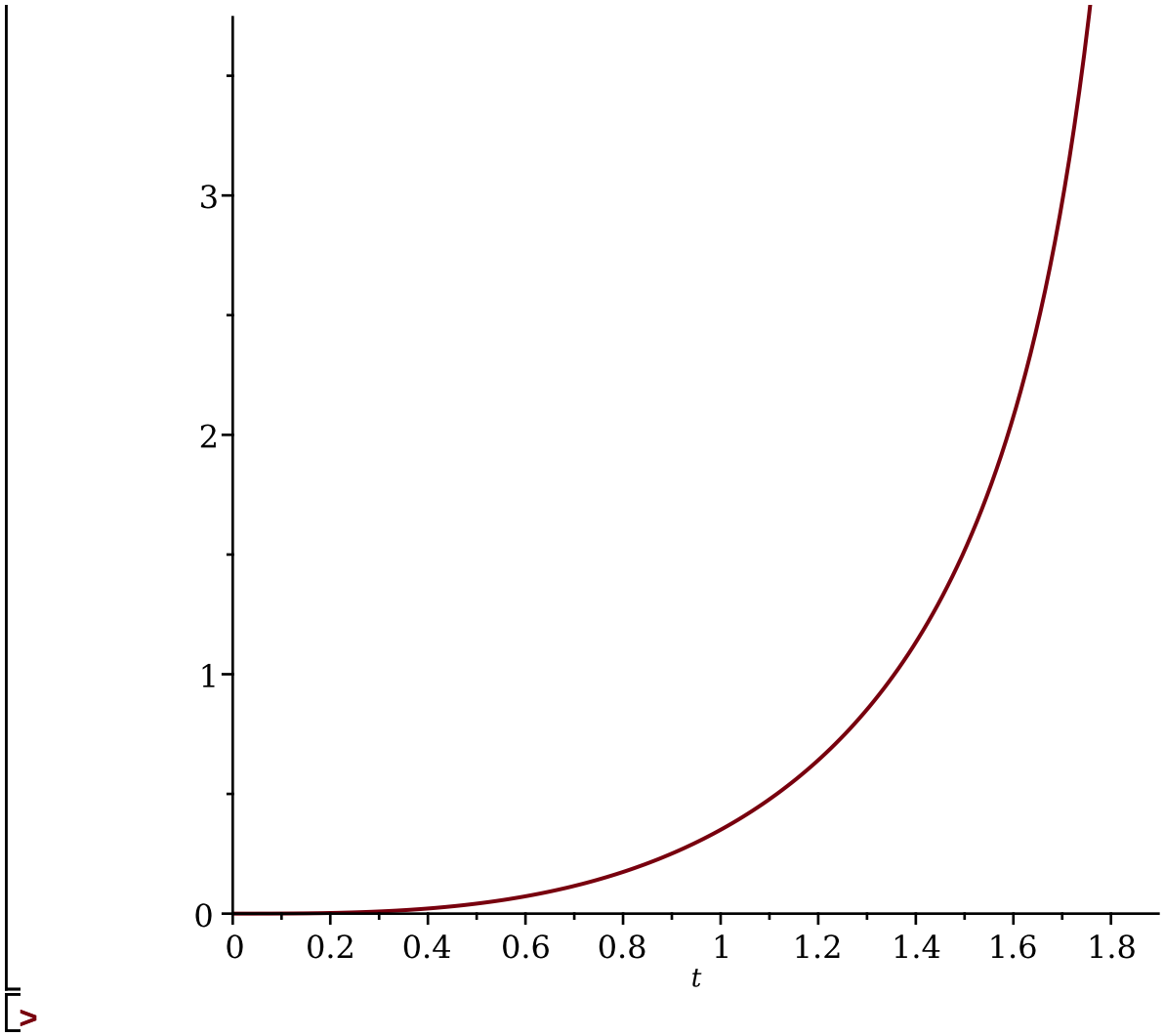
2.0, 23.42048639, 317.7224621, 294.3019757 **(73)**

```
> y='y'
```

$$23.42048639 = y \quad \textbf{(74)}$$

```
> points := [[n, psi(n)]$n = 1..20]: with(plots): pointplot(points, symbol
      = circle); plot(phi(t), t = 0..2)
```


20
15
10
5
0
2
4
6
8
10
12
14
16
18
20


3
2
1
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
t
v